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**STABILIZATION OF PERIODIC TRAJECTORIES
THROUGH RECEDING HORIZON
OPTIMIZATION OF SWITCHING TIMES**

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STABILIZATION OF PERIODIC TRAJECTORIES THROUGH RECEDING HORIZON OPTIMIZATION OF SWITCHING TIMES

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*"If I have seen farther than others,
it is because I stood on the shoulders of giants."*

— SIR ISAAC NEWTON

Resumo

O presente trabalho traz duas contribuições. Inicialmente determinam-se trajetórias periódicas do estado de um sistema dinâmico em tempo discreto, resultantes da aplicação de sequências de entrada bidirecionais do tipo liga/desliga sujeitas a restrições temporais de chaveamento. Neste sentido, a contribuição deste trabalho consiste em caracterizar sequências de entrada factíveis, bem como as trajetórias do estado associadas em termos de um conjunto de restrições lineares envolvendo variáveis inteiras. A segunda parte do trabalho consiste na formulação de uma lei de controle preditivo, fazendo uso de perturbações nos instantes de chaveamento, para levar o estado de um dado ponto inicial para uma trajetória alvo, selecionada através de algum critério de engenharia na etapa anterior. Para isto utiliza-se um procedimento de linearização para descrever a relação entre perturbações nos instantes de chaveamento e a trajetória do estado. Para testar as metodologias propostas apresentam-se dois exemplos numéricos. O primeiro exemplo utiliza um modelo do tipo integrador duplo, o segundo exemplo aborda um modelo do tipo integrador duplo com modo flexível. Para cada exemplo determina-se uma trajetória periódica a ser empregada como alvo, e então a lei de controle é usada para conduzir o estado ao alvo desejado.

Abstract

The present work conveys two contributions. Initially, from the application of bidirectional on/off input sequences subject to switching-time constraints, the periodic state trajectories of a discrete-time dynamic systems were determined. In this sense, the contribution of this work consists in characterizing feasible input sequences and associated state trajectories in terms of linear constraints involving integer variables. The second part of this work consists in the formulation of a predictive control law, making use of perturbations in the switching times, to drive the state from a given initial point to the target trajectory, properly selected by using some engineering criterion in the previous stage. For this reason a linearization procedure is derived to describe the relationship between the perturbations in the switching times and the state trajectory. To evaluate the proposed methods two numerical examples are presented. The first example concerns to a double integrator model. The second example brings a model of the double integrator with flexible mode. For each example a periodic trajectory used as a target is determined, and thus the control law is applied to drive the state to the desired target.

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List of Abbreviations and Acronyms

DC/DC	Direct current/ direct current
MPC	Model predictive control
SC	Switching constraints

List of Symbols

T_s	sampling period;
$k \in \mathbb{Z}$	current discrete time;
$u(\cdot) \in \{-1, 0, 1\}$	N -periodic input sequence;
$x(k) \in \mathbb{R}^n$	state vector;
$A \in \mathbb{R}^{n \times n}$	constant matrix;
$B \in \mathbb{R}^{n \times n}$	constant matrix;
$L_{on} \in \mathbb{N}$	minimum number of consecutive elements equal to either -1 or $+1$;
$L_{off} \in \mathbb{N}$	minimum number of consecutive elements equal to 0 ;
$k \bmod N$	remainder of the division of k by N ;
$u_0^{N-1}(\cdot) \in \{-1, 0, 1\}$	one period of the N -periodic sequence $u(\cdot)$
$\Delta u(\cdot) \in \{-1, 0, 1\}$	N -periodic sequence of increments;
$\Delta u_0^{N-1}(\cdot) \in \{-1, 0, 1\}$	one period of the N -periodic sequence $\Delta u(\cdot)$;
$b(k) \in \{0, 1\}$	binary variable associated to the feasibility problem for $L_{on} = L_{off}$;
$\Delta_1 u(k) \in \{0, 1\}$	component of the control increment associated to the feasibility problem for $L_{on} \neq L_{off}$;
$\Delta_2 u(k) \in \{-1, 0, 1\}$	component of the control increment associated to the feasibility problem for $L_{on} \neq L_{off}$;
$\Delta_3 u(k) \in \{-1, 0\}$	component of the control increment associated to the feasibility problem for $L_{on} \neq L_{off}$;
$\gamma(k) \in \{0, 1\}$	binary variable associated to the feasibility problem for $L_{on} \neq L_{off}$;
Δt_{ON}	minimum ON time;
Δt_{OFF}	minimum OFF time;
T	oscillation period;
$x_r(t) \in \mathbb{R}^n$	target trajectory;
t_{r_i}	i -th switching time;
$t_i \in \mathbb{R}$	i -th perturbed switching time;
$\delta t_i \in \mathbb{R}$	i -th switching-time perturbation;

$\doteq$	first-order approximation;
$\hat{x}(t_{r_{k+i}} + \hat{\delta}t_{k+1 k} + \cdots + \hat{\delta}t_{k+i k})$	predicted value of the state at the i -th perturbed switching time;
$\hat{\delta}t_{k+i k}^*$	optimized sequence of perturbations;
J	cost function;
Q, ρ	design weights;
$\mathcal{H}$	Hessian matrix;
$\mathcal{X}_F \subset \mathbb{R}^n$	feasibility region in the state space.

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1 Introduction

1.1 Motivation

Switched control systems have been the object of extensive research (HESPANHA, 2004), (SUN; GE, 2005), (ZHANG; GAO, 2010). They are used in a variety of engineering applications, including e.g. DC-DC converters (LOXTON *et al.*, 2009), vehicle suspension structures (DU *et al.*, 2014), and induction generators (TAHERZADEH *et al.*, 2017). The operation of such systems can involve transitions between two or more subsystems, such as, distinct circuit topologies in power converters (LOXTON *et al.*, 2009). The switched nature can also arise from the commutation among different controllers. Wu *et al.* (2013), for example, addressed a multi-proportional-integral-derivative controller tuning problem, in which the switching-time instants, the switching sequence, and the controller parameters should be determined for each subsystem. The control signal itself can have a switched nature, as for instance in Kienitz (2006), where a controller was designed to attain a chosen limit cycle.

Several techniques have been proposed to determine suitable switching times within an optimization framework. An algorithm based on gradient-descent was presented in Egerstedt *et al.* (2003) to minimize a cost functional with the switching times as control variables. The joint optimization of switching times and controller parameters was addressed in Wu *et al.* (2013) by introducing a penalty function in the cost functional. Liu *et al.* (2014) proposed a gradient-based algorithm to optimize the switching times and adjustable parameter values of the subsystem dynamics.

In this context, difficulties may arise if the optimization is subjected to constraints on the time interval between consecutive switching actions. For instance, these constraints must be addressed in the operation of distributed generation systems due to a minimum period of time required to turn on or shut down individual generators (TRECATE *et al.*, 2004; LARSEN *et al.*, 2013; LARSEN *et al.*, 2014; PARISIO *et al.*, 2014). The operation of power electronic systems may also be subject to switching-time constraints, owing to physical limitations of the semiconductor components (LOXTON *et al.*, 2009), as well as the loss of energy caused by excessive switching (ARNTZEN; MAKSIMOVIC, 1998). Another

possibility is to impose a switched control profile for engineering issues. For example, in the attitude control of rockets or satellites in the upper atmosphere, the thrusters should be fully opened in one of two possible directions or remain closed, for reliability reasons. This on/off actuation scheme is subject to switching-time constraints, owing to the time required to initialize or extinguish the flow/burn of propellant (ANTROPOV *et al.*, 1999; ROM; GANY, 1992), resulting in a persistent motion, which can become quasi-periodic or chaotic. The goal then may consist in achieving a stable periodic motion with suitable amplitude and frequency. A more detailed review of this topic is presented in the master's theses of Mesquita (2006) and Chagas (2007).

Optimal control methods which take into account switching-time constraints are in general composed by two stages. Firstly the original problem is converted into an equivalent constrained nonlinear optimization problem, parameterized by the switching times. The second stage consists in obtaining a solution through the use of proper numerical solvers (XU; ANTSAKLIS, 2004; LOXTON *et al.*, 2009; STELLATO *et al.*, 2017). As an example, Xu and Antsaklis (2004) proposed a method based on the solution of a two point boundary value differential algebraic equation composed by the state, costate, boundary, and continuity constraints. Loxton *et al.* (2009) introduced a set of auxiliary differential equations and applied a time-scaling transformation in the first stage, then the equivalent problem was solved by using conventional algorithms. Stellato *et al.* (2017) did a similar problem transformation in the first stage, and in the second stage each iteration of the proposed algorithm employed linearized dynamics in order to compute the gradient and Hessian of the cost function. However, in these cases there is no guarantee that the converted optimization problem will be convex, which brings difficulties to obtain a numerical solution.

The specific problem of attitude control through the use of thrusters with switching-time constraints was addressed in Kienitz and Bals (2005) through the use of a suitable set of modulation curves for a pulse modulator. In Kienitz (2006) a controller was designed to attain a chosen limit cycle with switching-time constraints explicitly taken into account by using ideas from relay control theory in the time domain. A model predictive control (MPC) approach was proposed in the doctorate thesis of Vieira (2013)¹ by using control actions in the set $\{-1, 0, 1\}$, where $+1$ and -1 correspond to fully opened thrusters (in the clockwise or counter-clockwise direction) and 0 corresponds to closed thrusters. Switching-time constraints were imposed by placing restrictions on the minimum number of consecutive equal elements in the discrete-time control sequence. Extensive numerical simulations showed that the system trajectory did converge to a limit cycle around the desired set point. However, the proposed formulation did not offer *a priori* guarantees of

¹With preliminary results reported in (VIEIRA *et al.*, 2008; VIEIRA *et al.*, 2009; VIEIRA *et al.*, 2010; VIEIRA *et al.*, 2011)

convergence, due to the lack of a method for finding periodical trajectories that could be used as feasible targets for the MPC controller.

1.2 Contributions of the present work

The present work addresses the control of systems with actuators subject to switching-time constraints, by following the approach employed in Vieira (2013), where the control actions are restricted to integer values in the set $\{-1, 0, 1\}$. In this context, the main contributions of the present work are:

- **Constraint formulation to characterize feasible periodic trajectories.** In fact, the formulation in Vieira (2013) was not tailored for periodic sequences, which hampered its use for the determination of feasible target trajectories for the closed-loop predictive controller. In this work, the switching-time and periodicity constraints are expressed as linear inequality and equality restrictions imposed on a period of the input sequence. Integer linear programming solvers can then be employed to obtain all feasible input sequences and associated state trajectories. The results may be used to choose a convenient trajectory in terms of engineering criteria such as fuel expense and/or amplitude of oscillation of certain variables;
- **Predictive control formulation employing perturbations on the switching times.** In Vieira (2013), the predictive control law was based on the optimization of the sequence of integer-valued control actions, which required the solution of a mixed-integer programming problem. Herein, the predictive controller acts through real-valued perturbations in the switching times, in order to drive the system state to a pre-selected periodic target trajectory. A linearization procedure is derived to describe the relation between the switching-time perturbations and the resulting state trajectory. As a result, the perturbations can be optimized by solving a standard quadratic programming problem.

1.3 Outline of the remaining chapters

The remainder of this text is organized as follows. Chapter 2 states the problem in more detail, introducing the adopted system model and the switching-time and periodicity constraints to be enforced. The main contribution of this Chapter consists in the proposed codification for the constraints. A double integrator and a two-disk-spring model are employed for illustration purposes. Chapter 3 presents the proposed predictive control formulation, which aims to drive the system state to a target periodic trajectory, through

perturbations in the switching times. Simulation results are presented. Finally, Chapter 4 brings some concluding remarks.

2 Determination of periodic trajectories of dynamic systems subject to switching input constraints

Part of the theoretical and simulation results presented in this chapter were published by Marcolino *et al.* (2017) as a research paper in the Journal of Optimization Theory and Applications.

Herein, the switching times will be assumed to be multiples of a time interval T_s . Under this simplifying assumption, the determination of feasible trajectories can be carried out in a discrete-time framework.

2.1 Statement of the problem

Let $u(\cdot) := \{u(k), k \in \mathbb{Z}\}$ be the input sequence to a system with discrete-time dynamics described by a state equation of the form

$$x(k+1) = Ax(k) + Bu(k) \quad (2.1)$$

where $x(k) \in \mathbb{R}^n$ is the state vector and $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times 1}$ are constant matrices with real entries. Sequence $u(\cdot)$ is assumed to be N -periodic, with values in the set $\{-1, 0, +1\}$, i.e.,

$$u(k) \in \{-1, 0, +1\}, \forall k \in \mathbb{Z} \quad (2.2)$$

$$u(k+N) = u(k), \forall k \in \mathbb{Z} \quad (2.3)$$

This sequence will be termed SC-admissible (switching-constraint admissible) for given integer values $L_{on} \geq 1$ and $L_{off} \geq 1$ if the following switching constraints are satisfied (VIEIRA, 2013):

- (SC1) direct transitions from -1 to $+1$ and vice versa are not allowed;
- (SC2) the number of consecutive elements equal to either -1 or $+1$ must be larger or equal to L_{on} ;
- (SC3) the number of consecutive elements equal to 0 must be larger or equal to L_{off} .

For illustration, consider the period-8 sequences depicted in Fig. 2.1 and assume that $L_{on} = L_{off} = 2$.

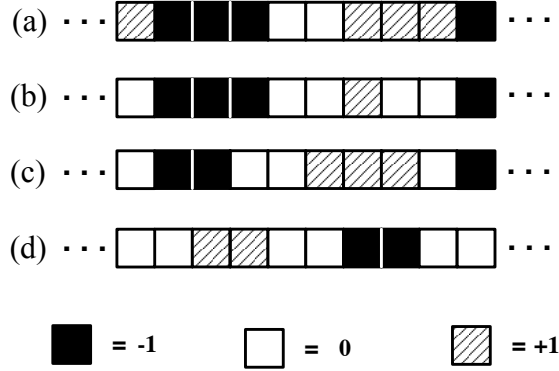


FIGURE 2.1 – Examples of period-8 input sequences.

Sequence (a) violates (SC1), because there are direct transitions from $+1$ to -1 . Sequence (b) violates (SC2) with $L_{on} = 2$, because there is a transition from 0 to $+1$ immediately followed by a transition from $+1$ to 0 . Sequence (c) violates (SC3) with $L_{off} = 2$, because there is a transition from $+1$ to 0 immediately followed by a transition from 0 to -1 . Finally, sequence (d) is SC-admissible, because it satisfies all the switching constraints.

In Vieira (2013), these switching constraints were formulated in terms of time increments on the input values, which were bounded by auxiliary integer-valued variables. Herein, this approach will also be employed. However, it is worth recalling that Vieira (2013) did not address the periodicity of the input and state sequences in a formal manner. In fact, the goal consisted of optimizing the input sequence over a moving time window within a predictive control formulation, without explicit periodicity constraints. In the present work, the coding of the switching constraints is aimed at formulating a feasibility problem, which will be solved in order to obtain an entire family of feasible periodic sequences.

In what follows, the sequence $u_0^{N-1}(\cdot) := \{u(k), k \in \{0, 1, \dots, N-1\}\}$ will represent one period of the infinite N -periodic sequence $u(\cdot)$, such that

$$u(k) = u_0^{N-1}(k \bmod N), \forall k \in \mathbb{Z} \quad (2.4)$$

where $(k \bmod N)$ denotes the remainder¹ of the division of k by N .

The problem under consideration consists of two parts:

- Codification of SC1, SC2 and SC3 in terms of constraints involving the N elements of the sequence $u_0^{N-1}(\cdot)$;
- Imposition of additional constraints to ensure the periodicity of the state trajectory $x(\cdot)$ resulting from the use of an SC-admissible sequence $u(\cdot)$. For this purpose, it is assumed that $x(\cdot)$ and $u(\cdot)$ are related through a state equation of the form (2.1).

The proposed codification of SC1, SC2 and SC3 will be presented in the next section, with the main results stated in Theorems 2.2.3 and 2.2.7 for $L_{on} = L_{off}$ and $L_{on} \neq L_{off}$, respectively. The periodicity constraint on the state trajectory can be imposed in a simple manner and will be presented in Proposition 2.2.10 at the end of the section. As will be shown, all the constraints can be formulated in a linear form, which enables the use of integer linear programming solvers to obtain all feasible sequences $u_0^{N-1}(\cdot)$.

2.2 Proposed codification for the switching constraints

Let $\Delta u(\cdot)$ be the N -periodic sequence of input increments defined as:

$$\Delta u(k) := u(k) - u(k-1), \forall k \in \mathbb{Z} \quad (2.5)$$

with corresponding period $\Delta u_0^{N-1}(\cdot) := \{\Delta u(k), k \in \{0, 1, \dots, N-1\}\}$, i.e.

$$\Delta u(k) = \Delta u_0^{N-1}(k \bmod N), \forall k \in \mathbb{Z} \quad (2.6)$$

In what follows, the switching constraints SC1, SC2, SC3 will be expressed in terms of $u_0^{N-1}(k)$ and $\Delta u_0^{N-1}(k)$, $k \in \{0, 1, \dots, N-1\}$.

Section 2.2.1 addresses the case $L_{on} = L_{off}$, in order to introduce the main ideas and notation employed in the proposed codification scheme. The case $L_{on} \neq L_{off}$ is presented in section 2.2.2.

¹For example, $(3 \bmod 5) = 3$, $(5 \bmod 5) = 0$ and $(7 \bmod 5) = 2$.

2.2.1 $L_{on} = L_{off}$

In this case, both L_{on} and L_{off} will be denoted by L , for simplicity.

Lemma 2.2.1. *If $u(\cdot)$ is SC-admissible, then $\exists b(k)$, $k \in \{0, 1, \dots, N-1\}$, such that:*

$$b(k) \in \{0, 1\}, \forall k \in \{0, 1, \dots, N-1\} \quad (2.7)$$

$$-b(k) \leq \Delta u_0^{N-1}(k) \leq b(k), \forall k \in \{0, 1, \dots, N-1\} \quad (2.8)$$

$$\sum_{k=i}^{i+L-1} b(k \bmod N) \leq 1, \forall i \in \{0, 1, \dots, N-1\} \quad (2.9)$$

Proof. Let $b(k)$ be chosen as

$$b(k) = |\Delta u_0^{N-1}(k)|, \forall k \in \{0, 1, \dots, N-1\} \quad (2.10)$$

With this choice, (2.8) is trivially satisfied. Under constraint SC1, direct transitions from $u(k-1) = -1$ to $u(k) = +1$ and vice-versa are not allowed. Therefore, in view of definitions (2.5) and (2.6), the possible values of $\Delta u(k)$ and $\Delta u_0^{N-1}(k)$ are in the set $\{-1, 0, +1\}$ and thus, the choice (2.10) is compatible with the constraint (2.7).

Constraints SC2 and SC3 impose that the number of consecutive elements with identical values be at least equal to L . Therefore, within a time window of length L , there can be at most one change in the value of $u(k)$, i.e. at most one element of $\Delta u(k)$ can be different from zero. As a consequence, since $\Delta u(k) \in \{-1, 0, +1\}$, the following expression holds:

$$\sum_{k=i}^{i+L-1} |\Delta u(k)| \leq 1, \forall i \in \mathbb{Z} \quad (2.11)$$

In view of (2.6) and (2.11), it follows that

$$\sum_{k=i}^{i+L-1} |\Delta u_0^{N-1}(k \bmod N)| \leq 1, \forall i \in \{0, 1, \dots, N-1\} \quad (2.12)$$

which shows that (2.9) is satisfied with the choice $b(k) = |\Delta u_0^{N-1}(k)|$. ■

Lemma 2.2.2. *If $u(\cdot)$ is not SC-admissible, then $\nexists b(k)$, $k \in \{0, 1, \dots, N-1\}$ that satisfy constraints (2.7), (2.8) and (2.9).*

Proof. If $u(\cdot)$ is not SC-admissible, then there is a violation of either SC1, SC2 or SC3. The violation of SC1 means that there is a direct transition from -1 to $+1$, or vice-versa,

at some time instant i , i.e. $u(i-1) = -1$, $u(i) = +1$ or $u(i-1) = +1$, $u(i) = -1$. In both cases, $|\Delta u(i)| = +2$, and thus $|\Delta u_0^{N-1}(k)| = +2$, with $k = (i \bmod N)$, which is not compatible with constraints (2.7) and (2.8). The violation of SC2 is related to a transition from either $u = +1$ to $u = 0$ or from $u = -1$ to $u = 0$. In the first case, the violation of SC2 means that there is a time instant i' and a time duration $L' < L$, such that $u(i'-1) = 0$, $u(i') = +1$ and $u(i'+L') = 0$. In this case, $\Delta u(i') = +1$ and $\Delta u(i'+L') = -1$, which implies that $\sum_{k'=i'}^{i'+L} |\Delta u(k')| > 1$. Owing to (2.6), it follows that $\sum_{k'=i'}^{i'+L} |\Delta u_0^{N-1}(k' \bmod N)| > 1$, i.e. $\sum_{k=i}^{i+L-1} |\Delta u_0^{N-1}(k \bmod N)| > 1$ for $i = i' \bmod N$. Therefore, if $b(k)$ is chosen in order to satisfy (2.8), then $\sum_{k=i}^{i+L-1} b(k \bmod N) > 1$, which is not compatible with (2.9). By using a similar reasoning, it can be shown that a violation of SC2 with $u(i'-1) = 0$ and $u(i'+L') = 0$ also renders the constraint infeasible. The analysis of a violation of SC3 is also similar. In this case, there are i' and $L' < L$ such that $u(i'-1) \neq 0$, $u(i') = 0$ and $u(i'+L') \neq 0$ implying $\sum_{k'=i'}^{i'+L} |\Delta u_0^{N-1}(k' \bmod N)| > 1$, i.e. $\sum_{k=i}^{i+L-1} b(k \bmod N) > 1$, with $i = i' \bmod N$, which results in the infeasibility of constraint (2.9). ■

Theorem 2.2.3. *An N -periodic sequence $u(\cdot)$ of integer values from the set $\{-1, 0, 1\}$ is SC-admissible if and only if $\exists b(k), k \in \{0, 1, \dots, N-1\}$ satisfying (2.7), (2.8) and (2.9).*

Proof. The necessity (only if) and sufficiency (if) follow from the contrapositives of Lemmas 2.2.1 and 2.2.2, respectively. ■

Remark 2.2.4. *In view of (2.4), (2.5), (2.6), $u_0^{N-1}(\cdot)$ and $\Delta u_0^{N-1}(\cdot)$ are related as*

$$\Delta u_0^{N-1}(k) = u_0^{N-1}(k) - u_0^{N-1}(k-1), \forall k \in \{1, 2, \dots, N-1\} \quad (2.13)$$

$$\Delta u_0^{N-1}(0) = u_0^{N-1}(0) - u_0^{N-1}(N-1) \quad (2.14)$$

where (2.14) follows from the N -periodicity of $u(\cdot)$. Therefore, the constraints (2.7), (2.8), (2.9) can be expressed in terms of the $2N$ variables $u_0^{N-1}(k)$, $b(k)$, $k \in \{0, 1, \dots, N-1\}$. An integer linear solver can then be used to find all feasible solutions of (2.7), (2.8), (2.9) involving these variables. The resulting solutions $u_0^{N-1}(\cdot)$ correspond to the desired SC-admissible sequences $u(\cdot)$. Indeed, Theorem 2.2.3 ensures that all sequences $u(\cdot)$ obtained in this manner will be SC-admissible and no SC-admissible sequence $u(\cdot)$ will be left out.

Example 2.2.1. Let $N = 8$ and $L = 2$. Fig. 2.2 depicts four sequences $u(\cdot)$ and the associated values of $\Delta u_0^7(k)$. In each case, the smallest values of $b(k)$ that satisfy (2.8) are also shown.

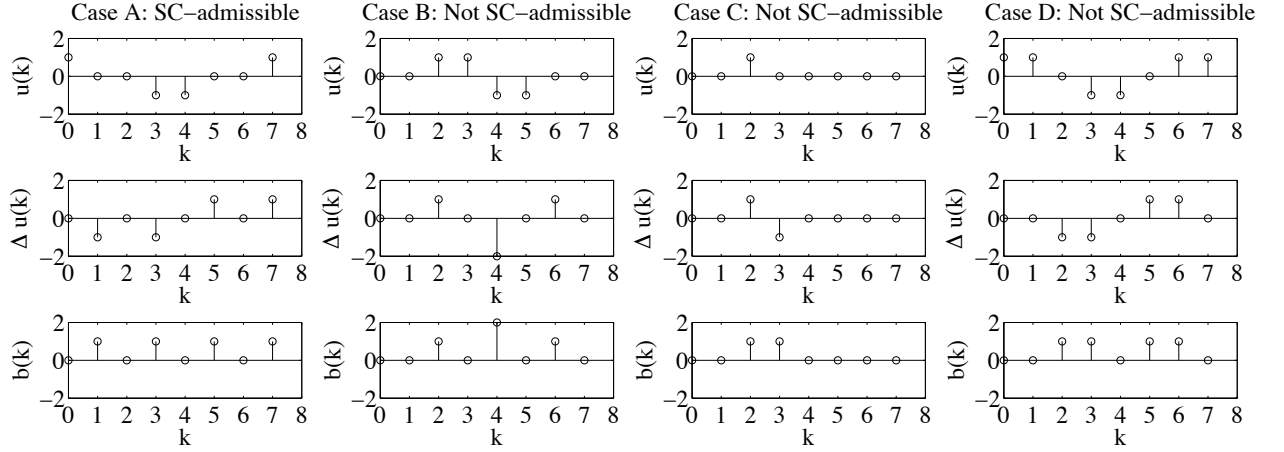


FIGURE 2.2 – Illustrative examples for $N = 8$ and $L = 2$.

Case A represents a SC-admissible sequence, for which the constraints (2.7), (2.8) and (2.9) are satisfied. In case B, there is a direct transition from $u(k-1) = +1$ to $u(k) = -1$ at time $k = 4$, which violates SC1. As a consequence, by enforcing (2.8) in the choice of $b(k)$, the constraints (2.7) and (2.9) were violated. Case C violates SC2 since $u(k-1) = 0$, $u(k) = +1$ and $u(k+1) = 0$ at time $k = 2$, having two transitions inside a window of length L . The constraints (2.7) and (2.8) are satisfied, but (2.9) is violated, because $b(k) + b(k+1) = +2$. Finally case D depicts a violation in SC3, as $u(k-1) = +1$, $u(k) = 0$ and $u(k+1) = -1$ at time $k = 2$, resulting in $b(k) + b(k+1) = +2$, i.e. the constraints (2.7) and (2.8) are satisfied, but (2.9) is not.

2.2.2 $L_{on} \neq L_{off}$

In order to enforce the switching constraints in the case $L_{on} \neq L_{off}$, the input increments $\Delta u(k)$ defined in (2.5) will be expressed as the sum of three components $\Delta_1 u(k)$, $\Delta_2 u(k)$ and $\Delta_3 u(k)$, with $\Delta_1 u(k) \in \{0, 1\}$, $\Delta_2 u(k) \in \{-1, 0, 1\}$, $\Delta_3 u(k) \in \{-1, 0\}$.

Lemma 2.2.5. *If $u(\cdot)$ is SC-admissible, then $\exists \gamma(k)$, $\Delta_1 u(k)$, $\Delta_2 u(k)$ and $\Delta_3 u(k)$ with $k \in \{0, 1, \dots, N-1\}$ such that:*

$$\Delta u_0^{N-1}(k) = \Delta_1 u(k) + \Delta_2 u(k) + \Delta_3 u(k) \quad (2.15)$$

$$\Delta_1 u(k) \in \{0, 1\} \quad (2.16)$$

$$\Delta_2 u(k) \in \{-1, 0, 1\} \quad (2.17)$$

$$\Delta_3 u(k) \in \{-1, 0\} \quad (2.18)$$

$$\Delta u_0^{N-1}(k) \in \{-1, 0, 1\} \quad (2.19)$$

$$\gamma(k) \in \{0, 1\} \quad (2.20)$$

$$-\gamma(k) \leq u_0^{N-1}(k) \leq \gamma(k) \quad (2.21)$$

$$\Delta_1 u(k) \leq \frac{1}{2} - \left[\frac{1}{2L_{on}} \sum_{j=-L_{on}}^{-1} u_0^{N-1}((k+j) \bmod N) \right] \quad (2.22)$$

$$|\Delta_2 u(k)| \leq 1 - \frac{1}{L_{off}} \sum_{j=-L_{off}}^{-1} \gamma((k+j) \bmod N) \quad (2.23)$$

$$-\Delta_3 u(k) \leq \frac{1}{2} + \left[\frac{1}{2L_{on}} \sum_{j=-L_{on}}^{-1} u_0^{N-1}((k+j) \bmod N) \right] \quad (2.24)$$

Proof. Let $\Delta_1 u(k)$, $\Delta_2 u(k)$ and $\Delta_3 u(k)$ be chosen as in Table 2.1, for each combination of $u(k-1)$ and $u(k)$ allowed by SC1. Moreover, let $\gamma(k)$ be chosen as

$$\gamma(k) = |u_0^{N-1}(k)|, \forall k \in \{0, 1, \dots, N-1\} \quad (2.25)$$

TABLE 2.1 – Feasible values for variables $\Delta_1 u(k)$, $\Delta_2 u(k)$, $\Delta_3 u(k)$ and $\Delta u_0^{N-1}(k)$ in Lemma 2.2.5 for $k \in \{0, 1, \dots, N-1\}$

$u(k-1)$	$u(k)$	$\Delta_1 u(k)$	$\Delta_2 u(k)$	$\Delta_3 u(k)$	$\Delta u_0^{N-1}(k)$
-1	0	1	0	0	1
0	-1	0	-1	0	-1
0	1	0	1	0	1
1	0	0	0	-1	-1
0	0	0	0	0	0
1	1	0	0	0	0
-1	-1	0	0	0	0

This choice of $\Delta_1 u(k)$, $\Delta_2 u(k)$, $\Delta_3 u(k)$, $\Delta u_0^{N-1}(k)$ and $\gamma(k)$ satisfies (2.15)-(2.21) trivially. The feasibility of (2.22) can be demonstrated by analyzing each of the possible cases $\Delta_1 u(k) = 0$ and $\Delta_1 u(k) = 1$. Indeed, the inequality $\sum_{j=-L_{on}}^{-1} u(k+j) \leq L_{on}$ holds for

any k , because $u(k+j) \in \{-1, 0, 1\}$. In view of (2.4), it follows that $\sum_{j=-L_{on}}^{-1} u_0^{N-1}((k+j) \bmod N) \leq L_{on}$ and thus the right-hand side of (2.22) will always be non-negative. Therefore (2.22) is trivially satisfied if $\Delta_1 u(k) = 0$. If $\Delta_1 u(k) = 1$, it follows from Table 1 that $u(k-1) = -1$ and $u(k) = 0$. In this case, the constraint SC2 ensures that $u(k+j) = -1$ for $j = -L_{on}, -L_{on} + 1, \dots, -1$. Therefore, $\sum_{j=-L_{on}}^{-1} u(k+j) = \sum_{j=-L_{on}}^{-1} u_0^{N-1}((k+j) \bmod N) = -L_{on}$ and thus the right-hand side of (2.22) will be equal to one. The feasibility of constraint (2.24) is demonstrated by using a similar reasoning, with $\Delta_3 u(k)$ instead of $\Delta_1 u(k)$.

Finally, the feasibility of (2.23) can be demonstrated by analyzing each possible value of $\Delta_2 u(k)$ in Table 1. If $\Delta_2 u(k) = 0$, (2.23) is trivially satisfied because (2.25) ensures that $\gamma(k) \leq 1$ and thus the right-hand side of (2.23) is always non-negative. If $\Delta_2 u(k) = \pm 1$, it follows from Table 1 that $u(k) = \pm 1$ and $u(k-1) = 0$. In this case, the constraint SC3 and the choice (2.25) ensure that $u(k+j) = u_0^{N-1}((k+j) \bmod N) = \gamma((k+j) \bmod N) = 0$ for $j = -L_{off}, -L_{off} + 1, \dots, -1$. Therefore, the right-hand side of (2.23) becomes equal to 1. ■

Lemma 2.2.6. *If $u(\cdot)$ is not SC-admissible, then $\nexists \gamma(k), \Delta_1 u(k), \Delta_2 u(k)$ and $\Delta_3 u(k)$ with $k \in \{0, 1, \dots, N-1\}$ that satisfy the constraints (2.15)-(2.24).*

Proof. If $u(\cdot)$ is not SC-admissible, then there is a violation of one or more of the constraints SC1, SC2 or SC3. The violation of SC1 means that there is a direct transition from -1 to $+1$, or vice-versa, at some time instant i , i.e. $u(i-1) = -1$ and $u(i) = +1$ or $u(i-1) = +1$ and $u(i) = -1$. In both cases $|\Delta u(i)| = +2$, and thus $|\Delta u_0^{N-1}(k)| = +2$, with $k = i \bmod N$, which conflicts with constraint (2.19).

The violation of SC2 is related to a transition from either $u = +1$ to $u = 0$ or from $u = -1$ to $u = 0$. In the first case, there are a time instant i' and a time duration $L' < L_{on}$, such that $u(i' - L' - 1) = 1$, $u(i' - 1) = 1$ and $u(i') = 0$. In this case, $\Delta u(i') = -1$, and thus from (2.6), $\Delta u(k')_0^{N-1} = -1$, with $k' = i' \bmod N$. In addition $\sum_{j=-L_{on}}^{-1} u(k'+j) < L_{on}$, and owing to (2.4) it follows that $\sum_{j=-L_{on}}^{-1} u_0^{N-1}((k'+j) \bmod N) < L_{on}$. Therefore, to ensure that (2.24) is satisfied, $\Delta_3 u(k')$ must be equal to zero. On the other hand the constraint (2.15) with $\Delta u_0^{N-1}(k') = -1$ imposes that $\Delta_2 u(k') = -1$ and $\Delta_1 u(k') = 0$. However, (2.23) cannot be satisfied with $|\Delta_2 u(k')| = +1$, because (2.21) with $u_0^{N-1}((k'-1) \bmod N) = 1$ imposes that $\gamma((k'-1) \bmod N) = 1$ and consequently $\sum_{j=-L_{off}}^{-1} \gamma((k'+j) \bmod N) \geq 1$. By using a similar reasoning, it can be shown that a violation of SC2 with $u(i' - 1) = -1$ and $u(i') = 0$ also renders the constraint infeasible.

The analysis of a violation of SC3 is also similar. In this case, there are i'' and $L'' < L_{off}$ such that $u(i'' - L'' - 1) \neq 0$, $u(i'' - 1) = 0$ and $u(i'') \neq 0$ which, in view of (2.21), implies $\sum_{j=-L_{off}}^{-1} \gamma((k''+j) \bmod N) > 0$, with $k'' = i'' \bmod N$. Therefore, $\Delta_2 u(k'')$ must be zero to ensure that (2.23) is satisfied. If $u(i'') = +1$ then from (2.6),

$\Delta u_0^{N-1}(k'') = +1$ which results in $\Delta_1 u(k'') = +1$ and $\Delta_3 u(k'') = 0$, owing to (2.15). Therefore, the constraint (2.22) is infeasible, because $\sum_{j=-L_{on}}^{-1} u_0^{N-1}((k'' + j) \bmod N) > -L_{on}$. If $u(i'') = -1$ then $\Delta u_0^{N-1}(k'') = -1$, $\Delta_1 u(k'') = 0$ and $\Delta_3 u(k'') = -1$ which results in the infeasibility of constraint (2.24). ■

Theorem 2.2.7. *An N -periodic sequence $u(\cdot)$ of integer values from the set $\{-1, 0, 1\}$ is SC-admissible if and only if $\exists \gamma(k)$, $\Delta_1 u(k)$, $\Delta_2 u(k)$ and $\Delta_3 u(k)$, $k \in \{0, 1, \dots, N-1\}$ satisfying (2.15)-(2.24).*

Proof. The necessity (only if) and sufficiency (if) follow from the contrapositives of Lemmas 2.2.5 and 2.2.6, respectively. ■

Remark 2.2.8. *From (2.15), it follows that*

$$\Delta_2 u(k) = \Delta u_0^{N-1}(k) - \Delta_1 u(k) - \Delta_3 u(k), \forall k \in \{0, 1, \dots, N-1\} \quad (2.26)$$

In view of (2.13), (2.14), and (2.26), the constraints (2.15)-(2.24) can be expressed in terms of $4N$ variables $u_0^{N-1}(k)$, $\Delta_1 u(k)$, $\Delta_3 u(k)$, and $\gamma(k)$, $k \in \{0, 1, \dots, N-1\}$. An integer linear solver can then be used to find all feasible solutions of (2.15)-(2.24) involving these variables. The resulting solutions $u_0^{N-1}(\cdot)$ correspond to the desired SC-admissible sequences $u(\cdot)$. Indeed, Theorem 2.2.7 ensures that all sequences $u(\cdot)$ obtained in this manner will be SC-admissible and no SC-admissible sequence $u(\cdot)$ will be left out.

Remark 2.2.9. *Theorem 2.2.7 also holds if $L_{on} = L_{off}$. However, in this case, the determination of all SC-admissible sequences can be more easily carried out by using Theorem 2.2.3, as it involves a smaller number of variables.*

It is worth noting that the characterization of SC-admissible sequences in Theorems 2.2.3 and 2.2.7 holds for any discrete dynamics, as the state equation (2.1) has not been employed so far. However, this equation will play a key role in a necessary and sufficient condition for the periodicity of the state trajectory $x(\cdot)$, as stated in Proposition 2.2.10 below.

Proposition 2.2.10. *If $u(\cdot)$ is an N -periodic sequence with period $u_0^{N-1}(\cdot)$, then the state trajectory $x(\cdot)$ starting from $x(0)$ is N -periodic if and only if the following condition holds:*

$$(I - A^N)x(0) = \begin{bmatrix} A^{N-1}B & A^{N-2}B & \dots & B \end{bmatrix} \begin{bmatrix} u_0^{N-1}(0) \\ u_0^{N-1}(1) \\ \vdots \\ u_0^{N-1}(N-1) \end{bmatrix} \quad (2.27)$$

Proof. (Necessity) If $x(\cdot)$ is an N -periodic sequence, then

$$x(k + N) = x(k), \forall k \geq 0 \quad (2.28)$$

Moreover, from the state equation (2.1), it follows that

$$x(N) = A^N x(0) + \begin{bmatrix} A^{N-1}B & A^{N-2}B & \dots & B \end{bmatrix} \begin{bmatrix} u_0^{N-1}(0) \\ u_0^{N-1}(1) \\ \vdots \\ u_0^{N-1}(N-1) \end{bmatrix} \quad (2.29)$$

Therefore, from (2.28) with $k = 0$ and (2.29), $u_0^{N-1}(\cdot)$ and $x(0)$ are related as in (2.27). ■

Proof. (Sufficiency) State equation (2.1) entails

$$x(k) = A^k x(0) + \begin{bmatrix} A^{k-1}B & A^{k-2}B & \dots & B \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(k-1) \end{bmatrix} \quad (2.30)$$

$$x(k + N) = A^{k+N} x(0) + \begin{bmatrix} A^{k+N-1}B & A^{k+N-2}B & \dots & B \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(k + N - 1) \end{bmatrix} \quad (2.31)$$

for any integer $k > 0$.

In view of (2.4) and (2.27), it follows that

$$x(0) = A^N x(0) + \begin{bmatrix} A^{N-1}B & A^{N-2}B & \dots & B \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix} \quad (2.32)$$

Replacing the right-hand side of (2.32) for $x(0)$ in (2.30), one obtains

$$\begin{aligned}
 x(k) = & A^{k+N}x(0) + \begin{bmatrix} A^{k+N-1}B & A^{k+N-2}B & \dots & A^k B \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix} \\
 & + \begin{bmatrix} A^{k-1}B & A^{k-2}B & \dots & B \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(k-1) \end{bmatrix}
 \end{aligned} \tag{2.33}$$

Owing to the N -periodicity of $u(\cdot)$, the following identity holds:

$$\begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(k-1) \end{bmatrix} = \begin{bmatrix} u(N) \\ u(N+1) \\ \vdots \\ u(k+N-1) \end{bmatrix} \tag{2.34}$$

Finally, from (2.33) and (2.34), it follows that

$$\begin{aligned}
 x(k) = & A^{k+N}x(0) + \begin{bmatrix} A^{k+N-1}B & A^{k+N-2}B & \dots & A^k B \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(N-1) \end{bmatrix} \\
 & + \begin{bmatrix} A^{k-1}B & A^{k-2}B & \dots & B \end{bmatrix} \begin{bmatrix} u(N) \\ u(N+1) \\ \vdots \\ u(k+N-1) \end{bmatrix} \\
 = & A^{k+N}x(0) + \begin{bmatrix} A^{k+N-1}B & A^{k+N-2}B & \dots & B \end{bmatrix} \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(k+N-1) \end{bmatrix}
 \end{aligned} \tag{2.35}$$

which corresponds to $x(k+N)$, as in (2.31). ■

If matrix A has one or more unit eigenvalues (i.e. if the system has integrating modes) then $(I - A^N)$ will be rank-deficient. In this case, the period sequence $u_0^{N-1}(\cdot)$ and the initial condition $x(0)$ will have to satisfy specific conditions derived from (2.27) as

discussed in Remarks 2.2.11 and 2.2.12 below.

Remark 2.2.11. *Constraint on the period sequence $u_0^{N-1}(\cdot)$. Let $p < n$ be the rank of matrix $(I - A^N)$. A singular value decomposition of $(I - A^N)$ can then be written as*

$$(I - A^N) = USV^T \quad (2.36)$$

where U , S and V are $n \times n$ matrices, such that $U^T = U^{-1}$, $V^T = V^{-1}$ and

$$S = \begin{bmatrix} s_1 & 0 & \cdots & 0 & \cdots & 0 \\ 0 & s_2 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & & \vdots \\ 0 & 0 & \cdots & s_p & \cdots & 0 \\ \vdots & \vdots & & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 \end{bmatrix} \quad (2.37)$$

with p nonzero singular values $s_1, \dots, s_p$. Now, let W and $\bar{u}$ be defined as

$$W = \begin{bmatrix} A^{N-1}B & A^{N-2}B & \cdots & B \end{bmatrix} \quad (2.38)$$

$$\bar{u} = \begin{bmatrix} u_0^{N-1}(0) \\ u_0^{N-1}(1) \\ \vdots \\ u_0^{N-1}(N-1) \end{bmatrix} \quad (2.39)$$

By using (2.36), (2.38) and (2.39), the identity (2.27) can be rewritten as

$$W\bar{u} = USV^T x(0) \quad (2.40)$$

After multiplying both sides of (2.40) by U^T , it follows that:

$$U^T W \bar{u} = SV^T x(0) \quad (2.41)$$

Let U and V be partitioned as

$$U = \begin{bmatrix} U_1 & U_2 \end{bmatrix} \quad (2.42)$$

$$V = \begin{bmatrix} V_1 & V_2 \end{bmatrix} \quad (2.43)$$

where $U_1, V_1 \in \mathbb{R}^{n \times p}$, and $U_2, V_2 \in \mathbb{R}^{n \times (n-p)}$. Then, in view of (2.37), (2.41) and (2.42), it follows that

$$U_2^T W \bar{u} = 0_{(n-p) \times 1} \quad (2.44)$$

which is a constraint that must be satisfied by the sequence $u_0^{N-1}(\cdot)$, in view of (2.39).

Remark 2.2.12. *Constraint on the initial condition $x(0)$. Assuming that $\bar{u}$ satisfies (2.44), a set of initial conditions $x(0)$ that satisfy (2.27) can be parameterized as follows.*

From (2.37), (2.41)-(2.43), it follows that:

$$S_p V_1^T x(0) = U_1^T W \bar{u} \quad (2.45)$$

where S_p is a $p \times p$ diagonal matrix, with the nonzero singular values $s_1, \dots, s_p$.

After multiplying both sides of (2.45) by S_p^{-1} , one obtains

$$V_1^T x(0) = S_p^{-1} U_1^T W \bar{u} \quad (2.46)$$

Therefore, the solutions for $x(0)$ will be of the form

$$x(0) = V_2 z + x_p(0) \quad (2.47)$$

where $z \in \mathbb{R}^{n-p}$ is a parameter vector and $x_p(0)$ is given by

$$x_p(0) = V_1 (V_1^T V_1)^{-1} S_p^{-1} U_1^T W \bar{u} \quad (2.48)$$

Indeed, from (2.47) and (2.48), one can write

$$V_1^T x(0) = V_1^T V_2 z + V_1^T V_1 (V_1^T V_1)^{-1} S_p^{-1} U_1^T W \bar{u} \quad (2.49)$$

Since the columns of $V = [V_1 \ V_2]$ are mutually orthogonal, one has $V_1^T V_2 = 0$, and thus (2.49) results in (2.46), for any parameter vector z .

2.3 Application examples

2.3.1 Double integrator

Consider a double integrator model of the form

$$\dot{x}(t) = A_c x(t) + B_c u(t) \quad (2.50)$$

where t is a continuous time variable and

$$A_c = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, B_c = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (2.51)$$

This model is used as a simplified description of the dynamics of mechanical systems with a single degree of freedom (RAO; BERNSTEIN, 2001). In this case, the state variables x_1, x_2 correspond to position and velocity, respectively, and the control input u is associated to a driving force or torque. Assuming that the control signal is kept constant between sampling times, the continuous-time model is discretized into the form (2.1) with

$$A = \exp(A_c T_s) = \begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix} \quad (2.52)$$

$$B = \int_0^{T_s} \exp(A_c \tau) B_c d\tau = \begin{bmatrix} T_s^2/2 \\ T_s \end{bmatrix} \quad (2.53)$$

where T_s is the sampling period (FRANKLIN *et al.*, 1998). Herein, a normalized value $T_s = 1$ s is employed, which leads to

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}. \quad (2.54)$$

In this case, one has

$$A^i = \begin{bmatrix} 1 & i \\ 0 & 1 \end{bmatrix}, i \in \mathbb{N} \quad (2.55)$$

and thus, the matrix W defined in (2.38) is given by

$$W = \begin{bmatrix} N - 0.5 & N - 1.5 & \cdots & 1.5 & 0.5 \\ 1 & 1 & \cdots & 1 & 1 \end{bmatrix} \quad (2.56)$$

Moreover, matrix $(I - A^N)$ is given by

$$(I - A^N) = \begin{bmatrix} 0 & -N \\ 0 & 0 \end{bmatrix} \quad (2.57)$$

which can be decomposed in the form (2.36) with

$$U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (2.58)$$

$$S = \begin{bmatrix} N & 0 \\ 0 & 0 \end{bmatrix} \quad (2.59)$$

$$V = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \quad (2.60)$$

As can be seen in (2.59), there is only one nonzero singular value $s_1 = N$. Therefore, matrices U , V can be partitioned as in (2.42), (2.43), with

$$U_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, U_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, V_1 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, V_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}. \quad (2.61)$$

As stated in Remark 2.2.11, from (2.39), (2.44), (2.56) and (2.61) the constraint on the period sequence $u_0^{N-1}(\cdot)$ is given by:

$$u_0^{N-1}(0) + u_0^{N-1}(1) + \cdots + u_0^{N-1}(N-2) + u_0^{N-1}(N-1) = 0 \quad (2.62)$$

Moreover, for a given period sequence $u_0^{N-1}(\cdot)$ subject to (2.62), the set of possible initial conditions $x(0)$ can be parameterized as in (2.47), (2.48) which results in

$$x_1(0) = -z \quad (2.63)$$

$$x_2(0) = \sum_{i=0}^{N-1} \left(\frac{-N + i + 0.5}{N} \right) u_0^{N-1}(i) \quad (2.64)$$

for any $z \in \mathbb{R}$.

Therefore, given a sequence $u_0^{N-1}(\cdot)$, different periodic state trajectories can be obtained by setting $x_2(0)$ as in (2.64) and varying the value of $x_1(0)$. It can be shown that all the resulting trajectories only differ by a translation along the x_1 axis².

Henceforth, a sequence $u_0^{N-1}(\cdot)$ will be said to be feasible if it satisfies (2.62) in addition

²Indeed, from (2.55) it follows that $x(i) = A^i x(0) + \sum_{k=0}^{i-1} A^k B u(i-k-1) = \begin{bmatrix} x_1(0) \\ 0 \end{bmatrix} + \begin{bmatrix} i x_2(0) \\ x_2(0) \end{bmatrix} + \sum_{k=0}^{i-1} A^k B u(i-k-1)$. Therefore, by changing $x_1(0)$, the entire state trajectory is shifted along the x_1 axis.

to the switching constraints SC1, SC2 and SC3. It is worth noting that $u_0^{N-1}(k) = 0, \forall k \in \{0, 1, \dots, N-1\}$ is a trivially feasible sequence and will not be considered in the results reported below.

For illustration, Figure 2.3 presents all the feasible sequences $u_0^{N-1}(\cdot)$ for the case $N = 5$ and $L_{on} = L_{off} = 1$. It is worth noting that any circular permutation of a feasible sequence is also feasible. Therefore, for ease of visualization, the 10 feasible sequences in this example will be presented in the form of two rings, as shown in the middle of Figure 2.3. Sequences (a)-(e) and (f)-(j) are formed by following the first and second rings in the clockwise direction, respectively, starting from different initial points.

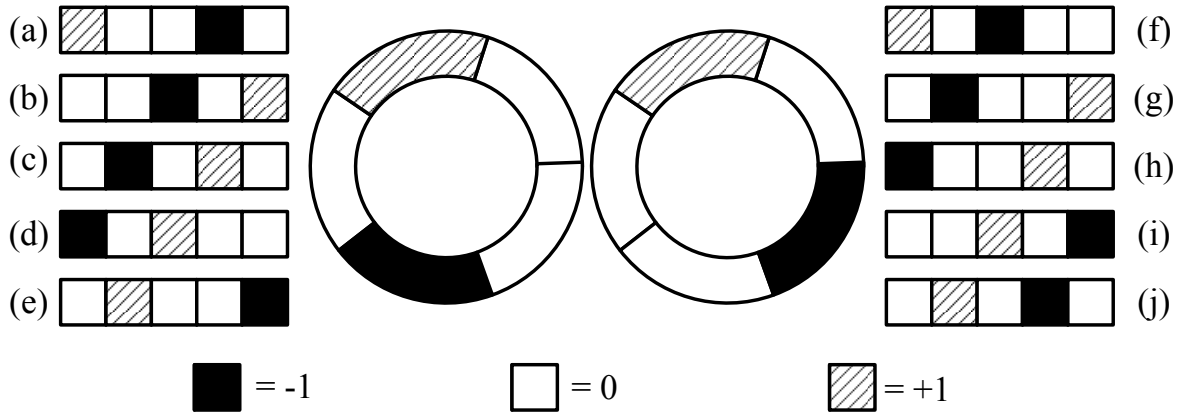


FIGURE 2.3 – Feasible sequences for $N = 5$ and $L_{on} = L_{off} = 1$

The CPLEX[®] software was used to obtain all feasible sequences by imposing the constraints in Theorem 2.2.3 (for $L_{on} = L_{off}$) and Theorem 2.2.7 (for $L_{on} \neq L_{off}$), in addition to (2.62), for different values of N , L_{on} and L_{off} . For the case where $L_{on} = L_{off}$, in light of Remark 2.2.4, $2N$ variables $u_0^{N-1}(k)$, $b(k)$, $k \in \{0, 1, \dots, N-1\}$ were employed, subject to (2.7), (2.8), (2.9), with $\Delta u_0^{N-1}(\cdot)$ given by (2.13) and (2.14). By using the *solution pool* feature of the CPLEX[®] software, all feasible solutions were generated and stored (CPLEX USER'S MANUAL, 2015). The same procedure was used for $L_{on} \neq L_{off}$: in view of Remark 2.2.8, $4N$ variables $u_0^{N-1}(k)$, $\Delta_1 u(k)$, $\Delta_3 u(k)$, and $\gamma(k)$, $k \in \{0, 1, \dots, N-1\}$ were employed, subject to (2.15)-(2.24), with $\Delta_2 u(k)$ given by (2.26). In a post-processing procedure, the number of associated rings (henceforth termed “feasible rings”) was determined by using the MATLAB[®] software.

The results for $L_{on} = L_{off} = L$ are shown in Figure 2.4. The number of feasible sequences and associated rings increases with the period length N . However, this number also depends on the parameter L . For $L = 3$, for example, the smallest period required to obtain non-trivial feasible sequences is $N = 12$. In this case, all the feasible sequences are associated to a single ring.

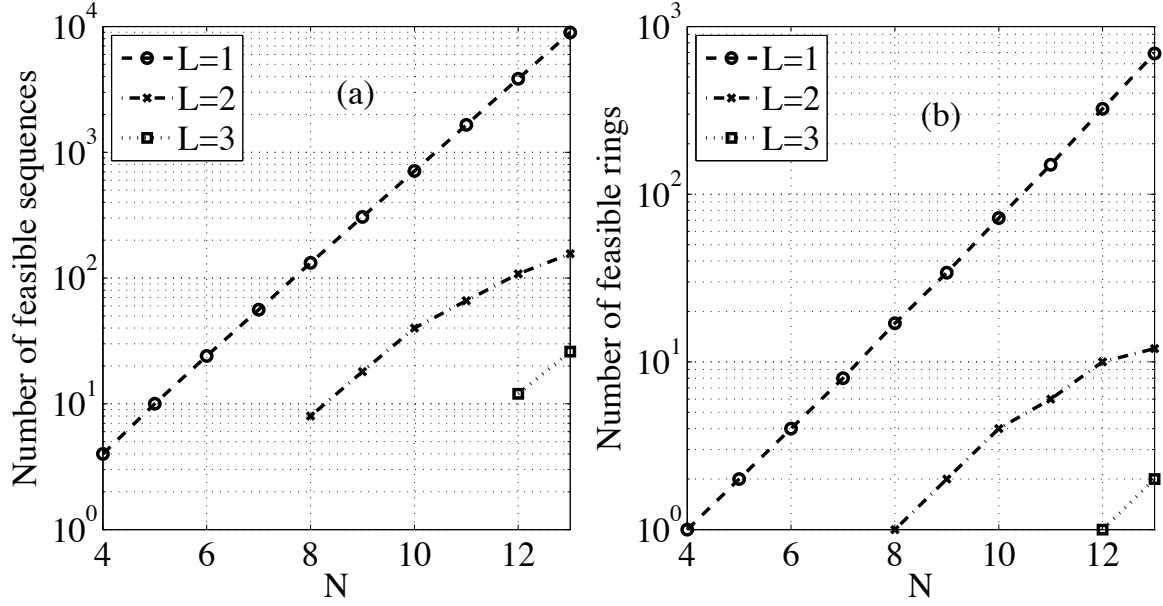


FIGURE 2.4 – (a) Number of feasible sequences and (b) number of associated rings as a function of the period N

The results for $L_{on} \neq L_{off}$ are shown in Table 2.2 for the case $N = 10$. For illustration, the nine feasible rings obtained for $L_{on} = 2$, $L_{off} = 1$ are presented in Figure 2.5. The associated state trajectories are depicted in Figure 2.6. These trajectories were generated by setting $x_2(0)$ as in (2.64) and adjusting $x_1(0)$ in order to obtain symmetry around $x_1 = 0$. The initial point adopted in each ring (i.e. the point corresponding to $u_0^{N-1}(0)$) is indicated by an asterisk $*$ in Figure 2.5. The corresponding initial point in the state trajectories is also indicated by a asterisk in Figure 2.6.

TABLE 2.2 – Number of feasible sequences and associated rings (in parenthesis) for $N = 10$ and different values of L_{on} , L_{off} .

L_{on}	L_{off}		
	1	2	3
1	710(72)	110(11)	40(4)
2	90(9)	40(4)	10(1)
3	40(4)	10(1)	0(0)

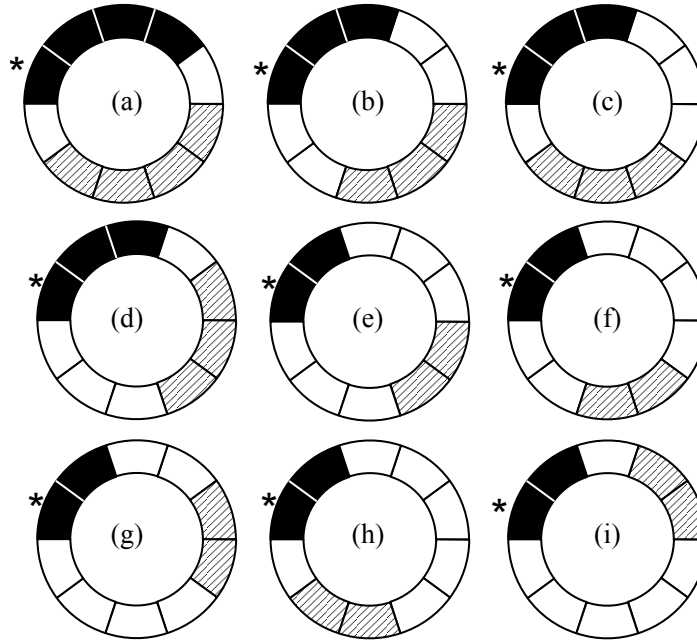


FIGURE 2.5 – Feasible rings for $N = 10$, $L_{on} = 2$ and $L_{off} = 1$. The asterisk * indicates the initial point adopted to generate the state trajectories in Figure 2.6.

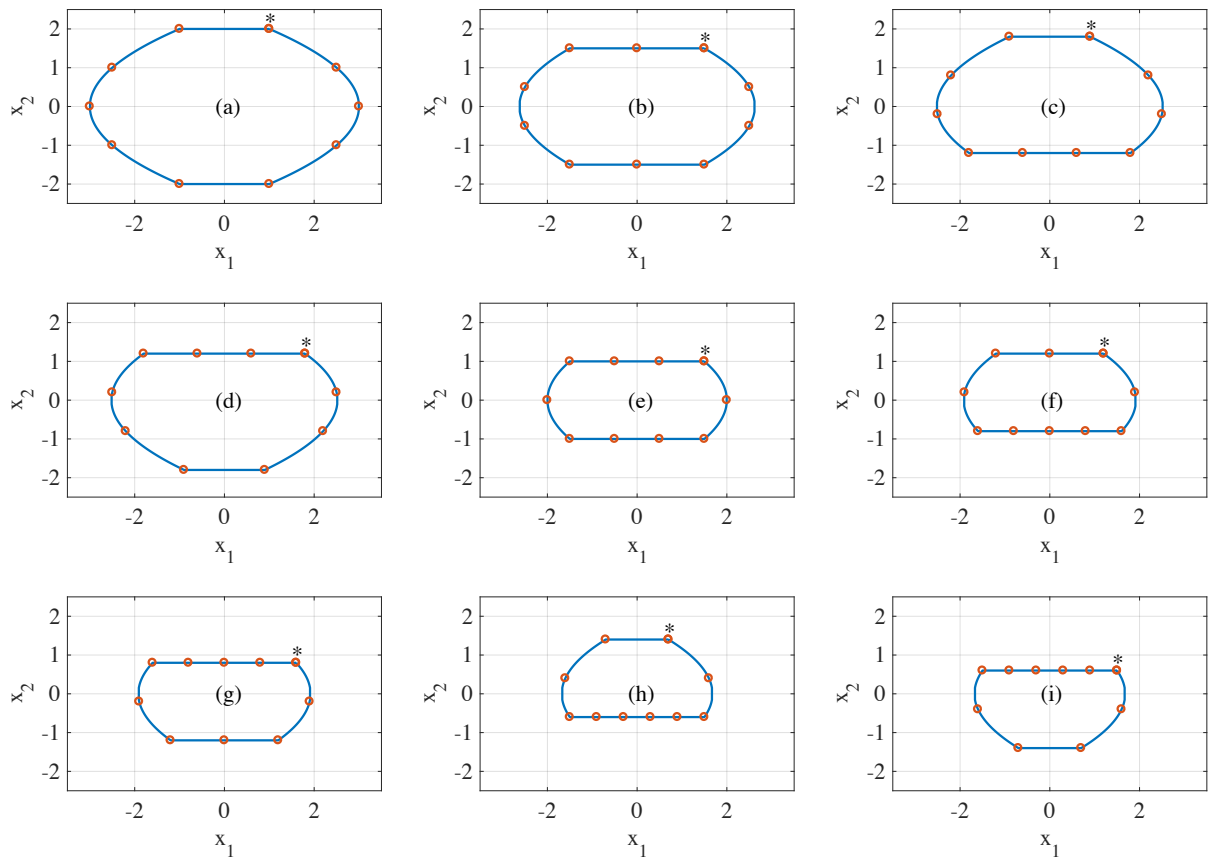


FIGURE 2.6 – Periodic state trajectories for the double integrator example

Engineering criteria can be used to choose the best trajectories among those depicted in Figure 2.6. As an example, the amplitude of the continuous-time trajectories along the x_1 axis is shown in Figure 2.7 in decreasing order. If small amplitudes for the position variable x_1 are required, the most appropriate choices will be trajectories (h) and (i).

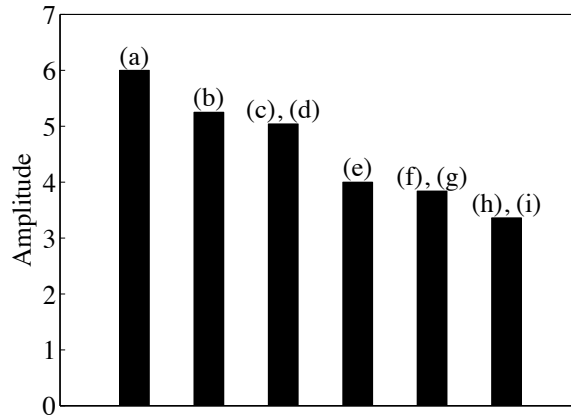


FIGURE 2.7 – Amplitudes of the state trajectories along the x_1 axis in Figure 2.6.

2.3.2 Two-disk-spring system

Consider the two-disk-spring system with thruster actuators depicted in Figure 2.8 below.

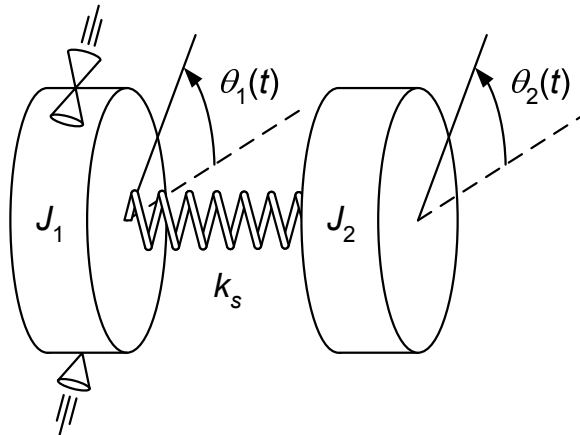


FIGURE 2.8 – Two-disk-spring system with thruster actuators.

The system dynamics can be described by a model of the form (2.50), with:

$$x = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}, A_c = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k_s/J_1 & k_s/J_1 & 0 & 0 \\ k_s/J_2 & -k_s/J_2 & 0 & 0 \end{bmatrix}, B_c = \begin{bmatrix} 0 \\ 0 \\ 1/J_1 \\ 0 \end{bmatrix}. \quad (2.65)$$

where k_s is the spring constant, J_1 and J_2 are the moment of inertia of the disks and the control input $u \in \{-1, 0, +1\}$ Nm corresponds to the torque generated by the thrusters. In this numerical example, the physical parameter values are assumed to be, $k_s = 0.2$ Nm/rad, $J_1 = 5$ kgm² and $J_2 = 4$ kgm², which result in the following matrices A_c and B_c :

$$A_c = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -0.04 & 0.04 & 0 & 0 \\ 0.05 & -0.05 & 0 & 0 \end{bmatrix}, B_c = \begin{bmatrix} 0 \\ 0 \\ 0.2 \\ 0 \end{bmatrix}. \quad (2.66)$$

Matrix A_c has eigenvalues at $\pm 0.3j$ and a double eigenvalue at 0. The sampling period was set to $T_s = 1$ s, which is approximately 20 times smaller than the natural oscillation period of the system, assuming that the control signal is kept constant between sampling times, the continuous-time model is discretized into the form (2.1), with:

$$A = \exp(A_c T_s) = \begin{bmatrix} 0.9801 & 0.0199 & 0.9934 & 0.0066 \\ 0.0248 & 0.9752 & 0.0083 & 0.9917 \\ -0.0394 & 0.0394 & 0.9801 & 0.0199 \\ 0.0493 & -0.0493 & 0.0248 & 0.9752 \end{bmatrix} \quad (2.67)$$

$$B = \int_0^{T_s} \exp(A_c \tau) B_c d\tau = \begin{bmatrix} 0.0997 \\ 0.0004 \\ 0.1987 \\ 0.0017 \end{bmatrix} \quad (2.68)$$

In what follows, the desired oscillation period will be set to $N = 10$. In view of (2.38), (2.67) and (2.68), the matrices W and $(I - A^N)$ are then given by:

$$W = \begin{bmatrix} 1.1404 & 1.1091 & 1.0630 & 0.9964 & 0.9054 & 0.7880 & 0.6449 & 0.4790 & 0.2951 & 0.0997 \\ 0.9495 & 0.7387 & 0.5462 & 0.3795 & 0.2433 & 0.1400 & 0.0688 & 0.0263 & 0.0062 & 0.0004 \\ 0.0263 & 0.0376 & 0.0555 & 0.0783 & 0.1041 & 0.1305 & 0.1552 & 0.1759 & 0.1909 & 0.1987 \\ 0.2171 & 0.2030 & 0.1806 & 0.1521 & 0.1199 & 0.0869 & 0.0560 & 0.0301 & 0.0114 & 0.0017 \end{bmatrix} \quad (2.69)$$

(2.70)

The matrix $(I - A^N)$ can be decomposed into the form (2.36) with

(2.71)

$$(2.72)$$

$$(2.73)$$

As can be seen, there are only $p = 3$ singular values different from zero in (2.72). Matrices U , V can be partitioned as in (2.42), (2.43), with

$$(2.74)$$

From (2.69) and (2.3.2), $U_2^T W$ is given by:

$$(2.75)$$

As stated in Remark 2.2.11, from (2.39), (2.44) and (2.75), the constraint on the period sequence $u_0^{N-1}(\cdot)$ is

$$0.1562 \sum_{i=0}^9 u_0^9(i) = 0 \quad (2.76)$$

which has the same form of constraint (2.62) in the first example. For a given period sequence $u_0^{N-1}(\cdot)$ subject to (2.76), the set of possible initial conditions $x(0)$ can be parameterized as in (2.47), (2.48) which results in

$$x(0) = \begin{bmatrix} -0.7071 \\ -0.7071 \\ 0 \\ 0 \end{bmatrix} z + \begin{bmatrix} 0.0350 & 0.0814 & 0.1204 & 0.1484 & 0.1631 & 0.1631 & 0.1484 & 0.1204 & 0.0814 & 0.0350 \\ -0.0350 & -0.0814 & -0.1204 & -0.1484 & -0.1631 & -0.1631 & -0.1484 & -0.1204 & -0.0814 & -0.0350 \\ -0.1543 & -0.1384 & -0.1190 & -0.0970 & -0.0732 & -0.0488 & -0.0250 & -0.0029 & 0.0164 & 0.0323 \\ -0.0446 & -0.0395 & -0.0387 & -0.0413 & -0.0460 & -0.0515 & -0.0562 & -0.0588 & -0.0580 & -0.0529 \end{bmatrix} \begin{bmatrix} u_0^9(0) \\ u_0^9(1) \\ u_0^9(2) \\ u_0^9(3) \\ u_0^9(4) \\ u_0^9(5) \\ u_0^9(6) \\ u_0^9(7) \\ u_0^9(8) \\ u_0^9(9) \end{bmatrix} \quad (2.77)$$

for any $z \in \mathbb{R}$.

Since the constraint on the period sequence remains the same as in the first example, by imposing the constraints in Theorem 2.2.3 (for $L_{on} = L_{off}$) and Theorem 2.2.7 (for $L_{on} \neq L_{off}$), the feasible input sequences remain the same. Therefore, for $L_{on} = 2$ and $L_{off} = 1$, the feasible rings are those presented in Fig. 2.5.

The associated state trajectories are depicted in Figure 2.9. These trajectories were generated adjusting z in order to obtain symmetry around $x_2 = 0$, assuming that the angular position of disk 2 should be regulated around zero.

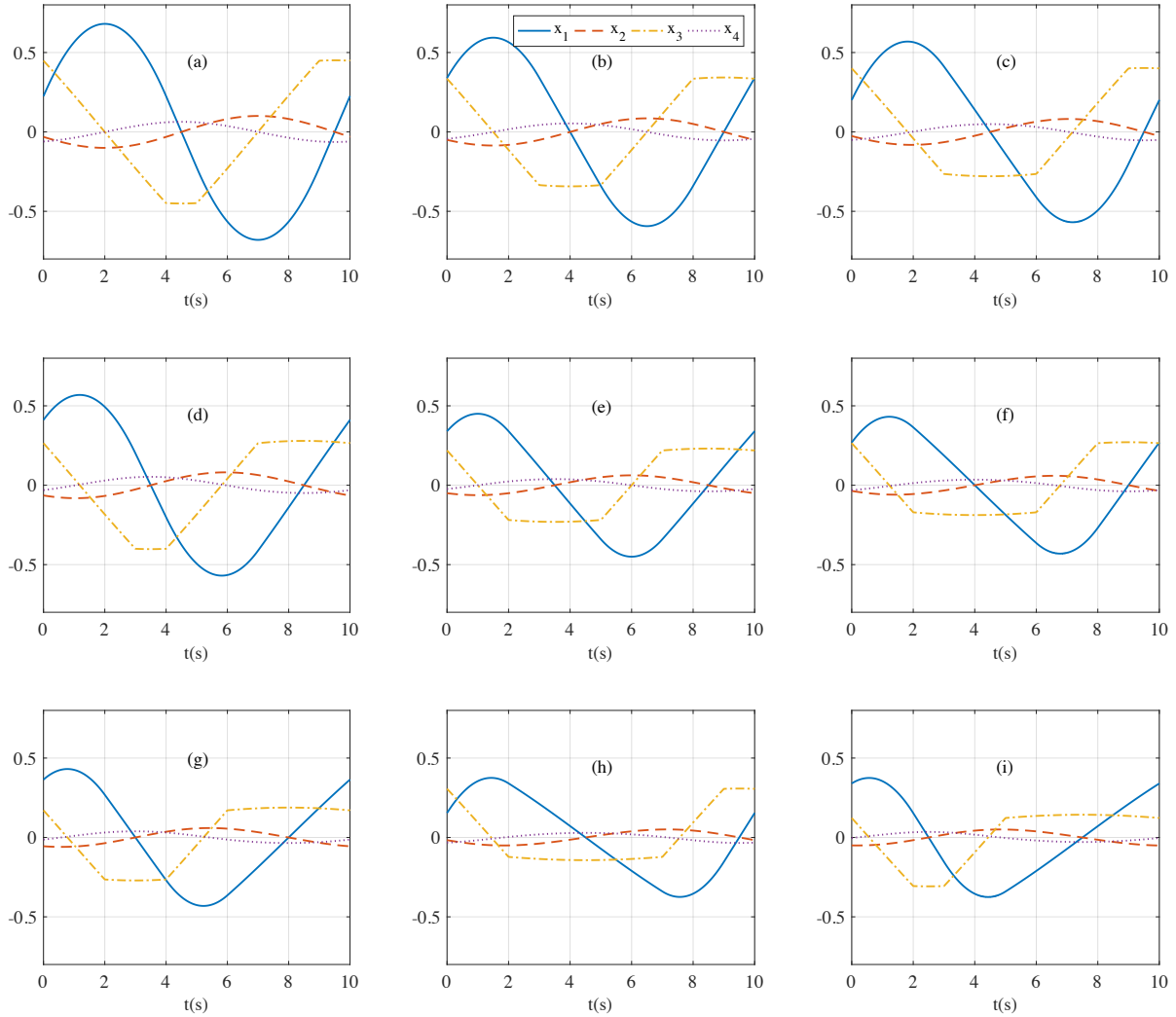


FIGURE 2.9 – Periodic state trajectories for the two-disk-spring example.

Engineering criteria can be used to choose the best trajectories among those depicted in Figure 2.9. For example, if the control action is aimed at regulating the position $x_2 = \theta_2$ of the second disk, a possible criterion could be the amplitude of the resulting x_2 trajectories which are depicted in Fig. 2.10 for better visualization. The continuous-time amplitude of these trajectories is shown in Figure 2.11 in decreasing order. If small amplitudes are desired, the most appropriate choices will be trajectories (h) and (i).

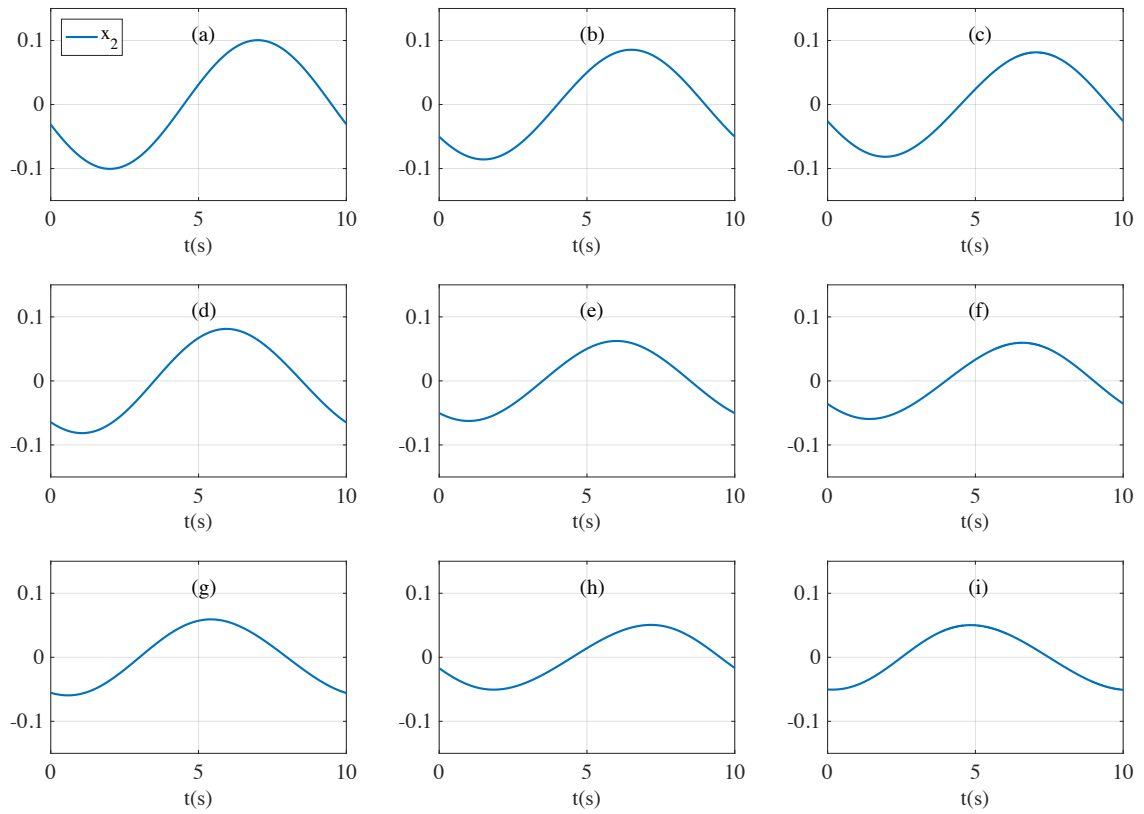


FIGURE 2.10 – Periodic x_2 signals.

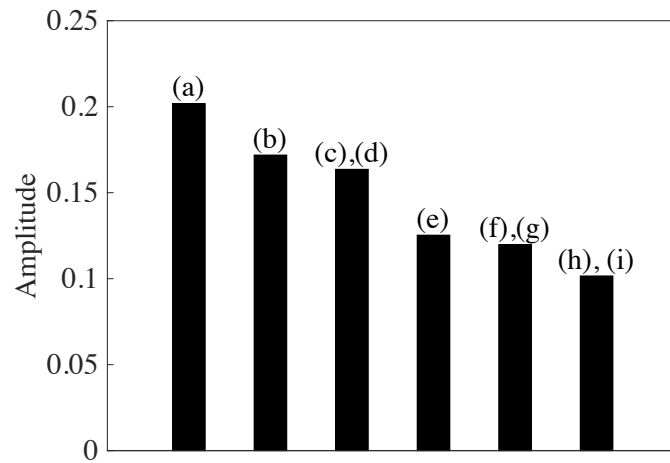


FIGURE 2.11 – Amplitude of the x_2 trajectories shown in Fig. 2.9.

2.4 Chapter conclusions

This chapter was concerned with the determination of periodic state trajectories arising from the application of bidirectional on/off inputs subject to switching-time constraints. For this purpose, the inputs were defined by values in the set $\{-1, 0, +1\}$ and the constraints were stated in terms of the minimum number of consecutive equal elements in the input sequence. The main contribution consisted in formulating such constraints in terms of linear inequalities that can be coded into an integer programming solver for the determination of all feasible input sequences with a given period. The periodicity of the state trajectory can then be imposed as an additional equality constraint.

Numerical examples using a double integrator and a two-disk-spring system were presented to illustrate the use of the proposed constraint formulation. The effect of the switching time constraints was clearly seen, in terms of a substantial reduction in the number of feasible input sequences as the constraints become more stringent. These examples also illustrated the possibility of selecting suitable state trajectories in terms of a given criterion, such as the amplitude of the oscillations in one of the state variables.

The next chapter is concerned with the actual design of a closed-loop controller aimed at steering the system state to the chosen periodic trajectory.

3 Predictive control for stabilization of periodic trajectories

3.1 Statement of the problem

Consider a plant with continuous-time dynamics described by a model of the form

$$\dot{x}(t) = A_c x(t) + B_c u(t) \quad (3.1)$$

where $t \in \mathbb{R}$ is the time variable, $x(t) \in \mathbb{R}^n$ is the state vector, $u(t) \in \{-1, 0, 1\}$ is the control input and $A_c \in \mathbb{R}^{n \times n}$, $B_c \in \mathbb{R}^{n \times 1}$ are matrices with constant coefficients. It is assumed that the control input is subject to switching-time constraints with minimum *on* and *off* times denoted by $\Delta t_{on} > 0$ and $\Delta t_{off} > 0$, respectively.

Let $x_r(t) \in \mathbb{R}^n$ denote a feasible target trajectory (in the sense defined in Chapter 2) associated to a piecewise constant input signal $u_r(t)$ of the form

$$u_r(t) = u_r(t_{r_i}), t_{r_i} \leq t < t_{r_{i+1}}, \forall i \in \mathbb{Z} \quad (3.2)$$

where t_{r_i} is the i th switching time. Both $x_r(t)$ and $u_r(t)$ are assumed to have a period T comprising N switching times.

Remark 3.1.1. *The framework presented in Chapter 2 can be employed to obtain feasible control and state trajectories with switching times t_{r_i} multiples of a time interval T_s . Herein, it is assumed that the switching-time constraints imposed in Chapter 2 were more restrictive than the actual actuator constraints, i.e. $L_{on}T_s > \Delta t_{on}$ and $L_{off}T_s > \Delta t_{off}$. Therefore, it will be possible to decrease the interval between the switching times, for control purposes.*

The problem addressed in this Chapter consists in determining perturbations in the N switching times over a period of oscillation, in order to drive the state from an initial point to the target trajectory. More precisely, after an initial switching time $t_0 = t_{r_0}$, the

perturbed switching times t_i will be given by

$$t_i - t_{i-1} = t_{r_i} - t_{r_{i-1}} + \delta t_i \quad (3.3)$$

where $\delta t_i \in \mathbb{R}$, $i = 1, 2, \dots, N$ are the manipulated variables in the control problem. The constraint $t_N = t_{r_N} = t_{r_0} + T$ is imposed to preserve the period of oscillation. The actual control signal $u(t)$ to be applied will be related to $u_r(t_{r_i})$ as

$$u(t) = u_r(t_{r_i}), \quad t_i \leq t < t_{i+1} \quad (3.4)$$

Figure 3.1 depicts the relation between signals $u_r(t)$ and $u(t)$.

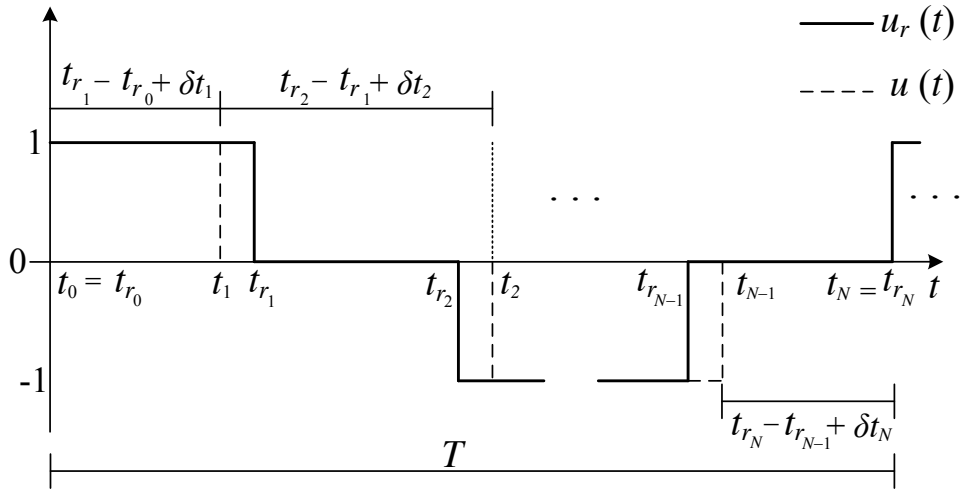


FIGURE 3.1 – Input signals $u_r(t)$ and $u(t)$.

It is worth noting that the switching-time perturbations affect the state trajectory in a nonlinear manner. In what follows, a linearized procedure will be derived to describe the relation between the perturbations $\delta t_1, \delta t_2, \dots, \delta t_N$ and the resulting state trajectory $x(t)$ at times $t_1, t_2, \dots, t_N$. Appropriate perturbations will then be determined through the solution of an optimal control problem.

3.2 Linearization procedure

Throughout this chapter, the following notation will be employed:

$$\Delta t_{r_i} = t_{r_i} - t_{r_{i-1}} \quad (3.5)$$

$$A_i \triangleq e^{A_c \Delta t_{r_i}} \quad (3.6)$$

$$B_i \triangleq \int_{t_{r_{i-1}}}^{t_{r_i}} e^{A_c(t_{r_i}-\tau)} B_c d\tau = \int_0^{\Delta t_{r_i}} e^{A_c \tau} B_c d\tau \quad (3.7)$$

for $i = 1, 2, \dots, N$. Moreover, the symbol $\doteq$ will be used to indicate a first-order approximation, as in Kirk (1970).

More specifically, bearing in mind that

$$e^{A_c \delta t} = I + A_c \delta t + \frac{(A_c \delta t)^2}{2!} + \dots \quad (3.8)$$

the following approximation will be employed, for small δt :

$$e^{A_c \delta t} \doteq I + A_c \delta t \quad (3.9)$$

In addition, consider the following integral:

$$\int_t^{t+\delta t} e^{A_c(t-\tau)} B_c d\tau = \int_{-\delta t}^0 e^{A_c \xi} B_c d\xi \quad (3.10)$$

where $t - \tau = \xi$ and $d\tau = -d\xi$. By using the approximation (3.9), one can write

$$\int_{-\delta t}^0 e^{A_c \xi} B_c d\xi \doteq \int_{-\delta t}^0 (I + A_c \xi) B_c d\xi = \left[\delta t I - \frac{A_c (\delta t)^2}{2} \right] B_c \doteq B_c \delta t \quad (3.11)$$

Remark 3.2.1. *The approximation error depends on the features of matrix A_c . For instance, if $A_c^2 = 0$, the relation (3.9) become exact.*

Theorem 3.2.2. *The relation between the switching time perturbations $\delta t_i \in \mathbb{R}$, $i \in \{1, 2, \dots, N\}$, and the resulting state trajectory starting from an initial condition $x(t_0)$ is given by the following first order approximation:*

$$x(t_i) \doteq G_{i,1} \delta t_1 + G_{i,2} \delta t_2 + \dots + G_{i,i} \delta t_i + f_i, \quad i = 1, 2, \dots, N \quad (3.12)$$

where

$$\begin{aligned} G_{1,1} &= A_c \left[A_1 x(t_0) + B_1 u_r(t_0) \right] + B_c u_r(t_0) \\ f_1 &= A_1 x(t_0) + B_1 u_r(t_0) \end{aligned} \quad (3.13)$$

and

$$\begin{aligned}
 G_{i,1} &= A_i G_{i-1,1} \\
 G_{i,2} &= A_i G_{i-1,2} \\
 &\vdots \\
 G_{i,i-1} &= A_i G_{i-1,i-1} \\
 G_{i,i} &= A_c A_i f_{i-1} + (A_c B_i + B_c) u_r(t_{r_{i-1}}) \\
 f_i &= A_i f_{i-1} + B_i u_r(t_{r_{i-1}})
 \end{aligned} \tag{3.14}$$

for $i = 2, 3, \dots, N$.

Proof. The proof of the theorem uses induction on i . Initially, it is worth recalling the solution for the state equation (3.1) from a given initial condition $x(t_0)$

$$x(t) = e^{A_c(t-t_0)}x(t_0) + \int_{t_0}^t e^{A_c(t-\tau)}B_c u(\tau) d\tau \tag{3.15}$$

From (3.4) and (3.15) with $t_0 = t_{r_0}$ and $t = t_1 = t_{r_1} + \delta t_1$, it follows that

$$x(t_1) = e^{A_c(t_{r_1}-t_{r_0}+\delta t_1)}x(t_0) + e^{A_c\delta t_1} \left[\int_{t_{r_0}}^{t_{r_1}+\delta t_1} e^{A_c(t_{r_1}-\tau)}B_c d\tau \right] u_r(t_{r_0}) \tag{3.16}$$

In view of (3.6) and (3.9), the first term on the right-hand side of (3.16) is approximated as

$$e^{A_c(t_{r_1}-t_{r_0}+\delta t_1)}x(t_0) = \underbrace{e^{A_c\delta t_1}}_{\doteq I+A_c\delta t_1} \underbrace{e^{A_c(t_{r_1}-t_{r_0})}}_{A_1}x(t_0) \doteq (I + A_c\delta t_1)A_1x(t_0) \tag{3.17}$$

Moreover, in view of (3.7) and (3.11), the integral term in (3.16) is approximated as

$$\begin{aligned}
 e^{A_c\delta t_1} \left[\int_{t_{r_0}}^{t_{r_1}+\delta t_1} e^{A_c(t_{r_1}-\tau)}B_c d\tau \right] u_r(t_{r_0}) &= \\
 \underbrace{e^{A_c\delta t_1}}_{\doteq I+A_c\delta t_1} \left[\underbrace{\int_{t_{r_0}}^{t_{r_1}} e^{A_c(t_{r_1}-\tau)}B_c d\tau}_{B_1} + \underbrace{\int_{t_{r_1}}^{t_{r_1}+\delta t_1} e^{A_c(t_{r_1}-\tau)}B_c d\tau}_{\doteq B_c\delta t_1} \right] u_r(t_{r_0}) & \\
 \doteq \left[B_1 + (B_c + A_c B_1)\delta t_1 \right] u_r(t_{r_0}) &
 \end{aligned} \tag{3.18}$$

where the second-order term $(\delta t_1)^2$ was neglected.

Therefore, in view of (3.17) and (3.18) the following approximation is derived from

(3.16), as follows

$$x(t_1) \doteq \underbrace{\left\{ A_c \left[A_1 x(t_0) + B_1 u_r(t_0) \right] + B_c u_r(t_0) \right\}}_{G_{1,1}} \delta t_1 + \underbrace{A_1 x(t_0) + B_1 u_r(t_0)}_{f_1} \quad (3.19)$$

which corresponds to (3.12) with $i = 1$ and $G_{1,1}$, f_1 as in (3.13).

Now, assume that (3.12) holds for $i = \bar{i}$, i.e.

$$x(t_{\bar{i}}) \doteq G_{\bar{i},1} \delta t_1 + G_{\bar{i},2} \delta t_2 + \cdots + G_{\bar{i},\bar{i}} \delta t_{\bar{i}} + f_{\bar{i}} \quad (3.20)$$

From (3.4) and (3.15) with $t_0 = t_{\bar{i}} = t_{r_{\bar{i}}} + \delta t_1 + \cdots + \delta t_{\bar{i}}$ and $t = t_{\bar{i}+1} = t_{r_{\bar{i}+1}} + \delta t_1 + \cdots + \delta t_{\bar{i}+1}$, it follows that

$$\begin{aligned} x(t_{\bar{i}+1}) &= e^{A_c(t_{r_{\bar{i}+1}} - t_{r_{\bar{i}}} + \delta t_{\bar{i}+1})} x(t_{\bar{i}}) + \left[\int_{t_{r_{\bar{i}}} + \delta t_1 + \cdots + \delta t_{\bar{i}}}^{t_{r_{\bar{i}+1}} + \delta t_1 + \cdots + \delta t_{\bar{i}+1}} e^{A_c(t_{r_{\bar{i}+1}} + \delta t_1 + \cdots + \delta t_{\bar{i}+1} - \tau)} B_c d\tau \right] u_r(t_{r_{\bar{i}}}) \\ &= \underbrace{e^{A_c \delta t_{\bar{i}+1}}}_{\doteq I + A_c \delta t_{\bar{i}+1}} \underbrace{e^{A_c(t_{r_{\bar{i}+1}} - t_{r_{\bar{i}}})}}_{A_{\bar{i}+1}} x(t_{\bar{i}}) + \\ &\quad \underbrace{e^{A_c \delta t_{\bar{i}+1}}}_{\doteq I + A_c \delta t_{\bar{i}+1}} \underbrace{\left[\int_{t_{r_{\bar{i}}} + \delta t_1 + \cdots + \delta t_{\bar{i}}}^{t_{r_{\bar{i}+1}} + \delta t_1 + \cdots + \delta t_{\bar{i}}} e^{A_c(t_{r_{\bar{i}+1}} + \delta t_1 + \cdots + \delta t_{\bar{i}} - \tau)} B_c d\tau \right]}_{B_{\bar{i}+1}} + \\ &\quad \underbrace{\left[\int_{t_{r_{\bar{i}+1}} + \delta t_1 + \cdots + \delta t_{\bar{i}}}^{t_{r_{\bar{i}+1}} + \delta t_1 + \cdots + \delta t_{\bar{i}+1}} e^{A_c(t_{r_{\bar{i}+1}} + \delta t_1 + \cdots + \delta t_{\bar{i}} - \tau)} B_c d\tau \right]}_{\doteq B_c \delta t_{\bar{i}+1}} u_r(t_{r_{\bar{i}}}) \\ &\doteq \left[A_c A_{\bar{i}+1} x(t_{\bar{i}}) + (A_c B_{\bar{i}+1} + B_c) u_r(t_{r_{\bar{i}}}) \right] \delta t_{\bar{i}+1} + A_{\bar{i}+1} x(t_{\bar{i}}) + B_{\bar{i}+1} u_r(t_{r_{\bar{i}}}) \end{aligned} \quad (3.21)$$

where the second-order term $(\delta t_{\bar{i}+1})^2$ was neglected.

Finally, by replacing (3.20) for $x(t_{\bar{i}})$ in (3.21) and neglecting second-order terms, the following first-order approximation can be derived:

$$\begin{aligned}
 x(t_{\bar{i}+1}) &\doteq \left[A_c A_{\bar{i}+1} \left(G_{\bar{i},1} \delta t_1 + \cdots + G_{\bar{i},\bar{i}} \delta t_{\bar{i}} + f_{\bar{i}} \right) + \left(A_c B_{\bar{i}+1} + B_c \right) u_r(t_{r_{\bar{i}}}) \right] \delta t_{\bar{i}+1} + \\
 &\quad \underbrace{A_{\bar{i}+1} f_{\bar{i}} + B_{\bar{i}+1} u_r(t_{r_{\bar{i}}})}_{f_{\bar{i}+1}} + A_{\bar{i}+1} \left(G_{\bar{i},1} \delta t_1 + \cdots + G_{\bar{i},\bar{i}} \delta t_{\bar{i}} \right) \doteq \\
 &\quad \underbrace{A_{\bar{i}+1} G_{\bar{i},1}}_{G_{\bar{i}+1,1}} \delta t_1 + \cdots + \underbrace{A_{\bar{i}+1} G_{\bar{i},\bar{i}}}_{G_{\bar{i}+1,\bar{i}}} \delta t_{\bar{i}} + \underbrace{\left[A_c A_{\bar{i}+1} f_{\bar{i}} + \left(A_c B_{\bar{i}+1} + B_c \right) u_r(t_{r_{\bar{i}}}) \right]}_{G_{\bar{i}+1,\bar{i}+1}} \delta t_{\bar{i}+1} + f_{\bar{i}+1} \\
 &\doteq G_{\bar{i}+1,1} \delta t_1 + \cdots + G_{\bar{i}+1,\bar{i}} \delta t_{\bar{i}} + G_{\bar{i}+1,\bar{i}+1} \delta t_{\bar{i}+1} + f_{\bar{i}+1}
 \end{aligned} \tag{3.22}$$

which corresponds to (3.12) with $i = \bar{i} + 1$ and $G_{i,1}, \dots, G_{i,i-1}, f_i$ as in (3.14) thus completing the proof by induction. $\blacksquare$

3.3 Optimal control problem

The optimal control problem addressed herein consists in determining a sequence $\{\hat{\delta} t_{k+i|k}, i = 1, 2, \dots, N\}$ of perturbations on the switching times of the nominal control signal, over one oscillation cycle starting at time $t = t_{r_k}$. The predicted value of the state at the i th perturbed switching time will be denoted by $\hat{x}(t_{r_{k+i}} + \hat{\delta} t_{k+1|k} + \cdots + \hat{\delta} t_{k+i|k})$. It is assumed that the state vector $x(t_{r_k})$ is known. The proposed control law consists of applying the optimized sequence of perturbations $\{\hat{\delta} t_{k+i|k}^*, i = 1, 2, \dots, N\}$ and repeating the entire procedure at time $t = t_{r_{k+N}}$, in a receding horizon manner (MACIEJOWSKI, 2002). In this sense, the proposed scheme is regarded as a predictive control formulation. In what follows, the cost function, the prediction equation and the constraints involved in the optimization problem will be presented. The term “optimization horizon” will be employed to indicate the time span over which the trajectory is optimized.

3.3.1 Cost function

Let $\hat{e}_{k+i|k}$ be the difference between the predicted state value and the target trajectory, defined as

$$\hat{e}_{k+i|k} = \hat{x}(t_{r_{k+i}} + \hat{\delta} t_{k+1|k} + \cdots + \hat{\delta} t_{k+i|k}) - x_r(t_{r_{k+i}}) \tag{3.23}$$

The optimization problem considered herein involves a cost function of the form

$$J = \sum_{i=1}^N \hat{e}_{k+i|k}^T Q \hat{e}_{k+i|k} + \rho \hat{\delta} t_{k+i|k}^2 \tag{3.24}$$

where $Q = Q^T > 0$, $\rho > 0$ are design weights. The use of this quadratic cost together with the linear prediction equation and the constraints to be presented below will lead to a quadratic programming problem. A numerical solution can then be obtained through the use of convex optimizers.

3.3.2 Prediction equation

By using the following notation:

$$\hat{\mathbf{x}} = \begin{bmatrix} \hat{x}(t_{r_{k+1}} + \hat{\delta}t_{k+1|k}) \\ \hat{x}(t_{r_{k+2}} + \hat{\delta}t_{k+1|k} + \hat{\delta}t_{k+2|k}) \\ \vdots \\ \hat{x}(t_{r_{k+N}} + \hat{\delta}t_{k+1|k} + \cdots + \hat{\delta}t_{k+N|k}) \end{bmatrix} \quad (3.25)$$

$$\hat{\delta\mathbf{t}} = \begin{bmatrix} \hat{\delta}t_{k+1|k} \\ \hat{\delta}t_{k+2|k} \\ \vdots \\ \hat{\delta}t_{k+N|k} \end{bmatrix} \quad (3.26)$$

$$\mathbf{f} = \begin{bmatrix} f_{k+1|k} \\ f_{k+2|k} \\ \vdots \\ f_{k+N|k} \end{bmatrix} \quad (3.27)$$

$$G = \begin{bmatrix} G_{1,1} & 0 & 0 & \cdots & 0 \\ G_{2,1} & G_{2,2} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ G_{N,1} & G_{N,2} & G_{N,3} & \cdots & G_{N,N} \end{bmatrix} \quad (3.28)$$

with G and $\mathbf{f}$ calculated as in Theorem 3.2.2, a **prediction equation** is written in the form

$$\hat{\mathbf{x}} = G\hat{\delta\mathbf{t}} + \mathbf{f} \quad (3.29)$$

Now, let $\mathbf{x}_r$ be the vector of target values defined as

$$\mathbf{x}_r = \begin{bmatrix} x_r(t_{r_{k+1}}) \\ x_r(t_{r_{k+2}}) \\ \vdots \\ x_r(t_{r_{k+N}}) \end{bmatrix} \quad (3.30)$$

In view of (3.23), (3.25), (3.26) and (3.30), the cost function (3.24) is rewritten as

$$J = (\hat{\mathbf{x}} - \mathbf{x}_r)^T \mathbf{Q} (\hat{\mathbf{x}} - \mathbf{x}_r) + \rho \hat{\boldsymbol{\delta t}}^T \hat{\boldsymbol{\delta t}} \quad (3.31)$$

where $\mathbf{Q} = \mathbf{Q}^T > 0$ is an $nN \times nN$ block-diagonal matrix defined as $\mathbf{Q} = \text{diag}(Q, Q, \dots, Q)$.

3.3.3 Constraints

The restrictions introduced herein are aimed at driving the predicted state to the target trajectory at the end of the optimization horizon, while respecting the switching-time and periodicity constraints. More specifically, the minimization of J defined in (3.31) will be subject to the following constraints

$$\sum_{i=1}^N \hat{\delta t}_{k+i|k} = 0 \quad (3.32)$$

$$\hat{x}(t_{r_{k+N}} + \hat{\delta t}_{k+1|k} + \dots + \hat{\delta t}_{k+N|k}) = x_r(t_{r_{k+N}}) \quad (3.33)$$

$$\hat{\delta t}_{k+i|k} \geq \delta t_{k+i,min} \quad (3.34)$$

where

$$\delta t_{k+i,min} = \begin{cases} -\Delta t_{r_{k+i}} + \Delta t_{off}, & \text{if } u_r(t_{r_{k+i-1}}) = 0 \\ -\Delta t_{r_{k+i}} + \Delta t_{on}, & \text{if } u_r(t_{r_{k+i-1}}) \neq 0 \end{cases} \quad (3.35)$$

for $i = 1, 2, \dots, N$.

The first constraint (3.32) is imposed to ensure that the period of the trajectory remains unaltered. The terminal constraint (3.33) is included to drive the predicted state to the target trajectory at the end of the oscillation cycle. Finally, inequality (3.34) enforces the switching-time constraints.

In view of (3.32), the optimization vector $\hat{\boldsymbol{\delta t}}$ is re-parameterized as

$$\hat{\boldsymbol{\delta t}} = H \hat{\mathbf{z}} \quad (3.36)$$

with

$$H = \begin{bmatrix} I_{(N-1) \times (N-1)} \\ -1_{N-1}^T \end{bmatrix}, \quad \hat{\mathbf{z}} = \begin{bmatrix} \hat{\delta t}_{k+1|k} \\ \vdots \\ \hat{\delta t}_{k+N-1|k} \end{bmatrix} \quad (3.37)$$

where $I_{(N-1) \times (N-1)}$ is an identity matrix with the indicated dimensions and 1_{N-1} is a column vector with $N - 1$ unit elements.

From (3.29) and (3.36), the prediction equation becomes

$$\hat{\mathbf{x}} = GH\hat{\mathbf{z}} + \mathbf{f} \quad (3.38)$$

The terminal constraint (3.33) can be rewritten as

$$E\hat{\mathbf{x}} = x_r(t_{r_{k+N}}) \quad (3.39)$$

where

$$E = \begin{bmatrix} 0_{n \times n(N-1)} & I_{n \times n} \end{bmatrix} \quad (3.40)$$

After replacing (3.38) for $\hat{\mathbf{x}}$, the constraint (3.39) becomes

$$EGH\hat{\mathbf{z}} = x_r(t_{r_{k+N}}) - E\mathbf{f} \quad (3.41)$$

where matrix EGH has dimensions $n \times (N - 1)$, as seen in (3.42):

$$\begin{aligned} EGH &= \begin{bmatrix} 0_{n \times n(N-1)} & I_{n \times n} \end{bmatrix} \begin{bmatrix} G_{1,1} & 0 & 0 & \cdots & 0 \\ G_{2,1} & G_{2,2} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ G_{N,1} & G_{N,2} & G_{N,3} & \cdots & G_{N,N} \end{bmatrix} \begin{bmatrix} I_{(N-1) \times (N-1)} \\ -1_{N-1}^T \end{bmatrix} \\ &= \begin{bmatrix} G_{N,1} - G_{N,N} & G_{N,2} - G_{N,N} & \cdots & G_{N,N-1} - G_{N,N} \end{bmatrix} \end{aligned} \quad (3.42)$$

If EGH has full row rank, (3.41) is taken into account by parameterizing $\hat{\mathbf{z}}$ as follows

$$\hat{\mathbf{z}} = F\hat{\boldsymbol{\theta}} + \mathbf{z}_0 \quad (3.43)$$

where $\hat{\boldsymbol{\theta}} \in \mathbb{R}^{N-1-n}$ is a new vector to be optimized, F is a matrix with $(N - 1 - n)$ linearly independent columns forming a basis for the null space of EGH , and $\mathbf{z}_0$ is any particular solution of (3.41) (GOULD *et al.*, 2001). Under the assumption that EGH is of full row rank, a solution $\mathbf{z}_0$ is given by

$$\mathbf{z}_0 = (EGH)^T \left(EGH(EGH)^T \right)^{-1} [x_r(t_{r_{k+N}}) - E\mathbf{f}] \quad (3.44)$$

After replacing (3.43) for $\hat{\mathbf{z}}$ in (3.36) and (3.38), it follows that

$$\hat{\delta t} = H(F\hat{\boldsymbol{\theta}} + \mathbf{z}_0) \quad (3.45)$$

$$\hat{\mathbf{x}} = GHF\hat{\boldsymbol{\theta}} + GH\mathbf{z}_0 + \mathbf{f} \quad (3.46)$$

and thus the cost function (3.31) rewritten in terms of $\hat{\boldsymbol{\theta}}$ as

$$\begin{aligned} J &= (GHF\hat{\boldsymbol{\theta}} + GH\mathbf{z}_0 + \mathbf{f} - \mathbf{x}_r)^T \mathbf{Q} (GHF\hat{\boldsymbol{\theta}} + GH\mathbf{z}_0 + \mathbf{f} - \mathbf{x}_r) + \rho (F\hat{\boldsymbol{\theta}} + \mathbf{z}_0)^T H^T H (F\hat{\boldsymbol{\theta}} + \mathbf{z}_0) \\ &= \hat{\boldsymbol{\theta}}^T \left[F^T H^T (G^T \mathbf{Q} G + \rho I) H F \right] \hat{\boldsymbol{\theta}} + 2 \left[(GH\mathbf{z}_0 + \mathbf{f} - \mathbf{x}_r)^T \mathbf{Q} G + \rho \mathbf{z}_0^T H^T \right] H F \hat{\boldsymbol{\theta}} + c \end{aligned} \quad (3.47)$$

where constant terms not related to $\hat{\boldsymbol{\theta}}$ are lumped into c .

J is a quadratic function of $\hat{\boldsymbol{\theta}}$, with Hessian matrix $\mathcal{H}$ given by

$$\mathcal{H} = 2F^T H^T (G^T \mathbf{Q} G + \rho I) H F \quad (3.48)$$

Remark 3.3.1. Assuming that the cost function weights are such that $\mathbf{Q} > 0$ and $\rho > 0$, the Hessian matrix $\mathcal{H}$ will be positive definite if and only if the product $HF \in \mathbb{R}^{N \times (N-1-n)}$ has full column rank. From (3.37), H has full column rank and thus $\text{rank}(HF) = \text{rank}(F)$ (BERNSTEIN, 2005). Moreover, $\text{rank}(F) = N - 1 - n$, because the columns of F are linearly independent. Therefore, it is concluded that HF has full column rank and thus $\mathcal{H}$ is positive definite.

Finally, in view of (3.45) the inequality constraints (3.34) can be written in terms of $\hat{\boldsymbol{\theta}}$ as

$$HF\hat{\boldsymbol{\theta}} \geq \delta t_{\min} - H\mathbf{z}_0 \quad (3.49)$$

where $\mathbf{z}_0$ is given by (3.44) and

$$\delta t_{\min} = \begin{bmatrix} \delta t_{k+1, \min} \\ \delta t_{k+2, \min} \\ \vdots \\ \delta t_{k+N, \min} \end{bmatrix} \quad (3.50)$$

The minimization, in terms of $\hat{\boldsymbol{\theta}}$, of J given in (3.47), subject to (3.49) is a quadratic programming problem (MACIEJOWSKI, 2002), which can be solved by using suitable numerical solvers.

3.3.4 Feasibility region

The feasibility region $\mathcal{X}_F \subset \mathbb{R}^n$ is defined as the set of initial states $x(t_0)$ for which the constraint (3.49) is feasible. The characterization of $\mathcal{X}_F$ can be carried out by using computational geometry operations. For this purpose, the following propositions will be initially established.

Proposition 3.3.2. *With E given in (3.40), the following identity holds:*

$$\begin{aligned} \hat{x}(t_{r_N} + \hat{\delta}t_{1|0} + \cdots + \hat{\delta}t_{N|0}) &= EG\hat{\delta}t + e^{A_c T}x(t_0) \\ &+ \begin{bmatrix} A_N A_{N-1} \cdots A_2 B_1 & A_N A_{N-1} \cdots A_3 B_2 & \cdots & B_N \end{bmatrix} \begin{bmatrix} u_r(t_{r_0}) \\ u_r(t_{r_1}) \\ \vdots \\ u_r(t_{r_{N-1}}) \end{bmatrix} \end{aligned} \quad (3.51)$$

Proof. Let $k = 0$. From the prediction equation (3.29), it follows that

$$\hat{x}(t_{r_N} + \hat{\delta}t_{1|0} + \cdots + \hat{\delta}t_{N|0}) = EG\hat{\delta}t + Ef \quad (3.52)$$

with Ef corresponding to $f_{N|0}$ in (3.27). By using the recursive formula for f_i presented in Theorem 3.2.2, one can write

$$\begin{aligned} f_1|0 &= A_1 x(t_0) + B_1 u_r(t_0) \\ f_2|0 &= A_2 f_1|0 + B_2 u_r(t_{r_1}) = A_2 A_1 x(t_0) + A_2 B_1 u_r(t_0) + B_2 u_r(t_{r_1}) \\ &\vdots \\ f_{N|0} &= A_N A_{N-1} \cdots A_1 x(t_0) + \cdots + A_N A_{N-1} \cdots A_2 B_1 u_r(t_0) + \\ &\quad + A_N A_{N-1} \cdots A_3 B_2 u_r(t_{r_1}) + \cdots + B_N u_r(t_{r_{N-1}}) \\ &= A_N A_{N-1} \cdots A_1 x(t_0) + \begin{bmatrix} A_N A_{N-1} \cdots A_2 B_1 & A_N A_{N-1} \cdots A_3 B_2 & \cdots & B_N \end{bmatrix} \begin{bmatrix} u_r(t_{r_0}) \\ u_r(t_{r_1}) \\ \vdots \\ u_r(t_{r_{N-1}}) \end{bmatrix} \end{aligned} \quad (3.53)$$

Moreover,

$$A_N A_{N-1} \cdots A_1 = e^{A_c \Delta t_{r_N}} e^{A_c \Delta t_{r_{N-1}}} \cdots e^{A_c \Delta t_{r_1}} = e^{A_c (\Delta t_{r_N} + \Delta t_{r_{N-1}} + \cdots + \Delta t_{r_1})} = e^{A_c T} \quad (3.54)$$

which completes the proof. ■

Proposition 3.3.3. *Under the constraint (3.32), the matrix EG in (3.51) does not depend on $x(t_0)$.*

Proof. If (3.32) is enforced, the following identity will hold:

$$\hat{x}(t_{r_N} + \hat{\delta}t_{1|0} + \cdots + \hat{\delta}t_{N|0}) = \hat{x}(t_{r_N}) \quad (3.55)$$

and thus (3.51) can be rewritten as

$$\hat{x}(t_{r_N}) - e^{A_c T} x(t_0) = EG\hat{\delta}t + \begin{bmatrix} A_N A_{N-1} \cdots A_2 B_1 & A_N A_{N-1} \cdots A_3 B_2 & \cdots & B_N \end{bmatrix} \begin{bmatrix} u_r(t_{r_0}) \\ u_r(t_{r_1}) \\ \vdots \\ u_r(t_{r_{N-1}}) \end{bmatrix} \quad (3.56)$$

Moreover, in view of (3.15), one can write

$$x(t_{r_N}) = e^{A_c(t_{r_N}-t_{r_0})} x(t_0) + \int_{t_{r_0}}^{t_{r_N}} e^{A_c(t_{r_N}-\tau)} B_c u(\tau) d\tau \quad (3.57)$$

i.e.

$$x(t_{r_N}) - e^{A_c T} x(t_0) = \int_{t_{r_0}}^{t_{r_N}} e^{A_c(t_{r_N}-\tau)} B_c u(\tau) d\tau \quad (3.58)$$

Since $\hat{x}(t_{r_N}) \doteq x(t_{r_N})$, the right-hand side of (3.56) must be a first-order approximation for the right-hand side of (3.58), which does not depend on $x(t_0)$. Therefore, it can be concluded that the term $EG\hat{\delta}t$ in (3.56) does not depend on $x(t_0)$. ■

In light of Propositions 3.3.2 and 3.3.3, one may conclude that z_0 in (3.44) can also be written as

$$z_0 = M [x_r(t_{r_N}) - \Phi x(t_0) - f_{Nu}] \quad (3.59)$$

where

$$M = (EGH)^T \left((EGH)(EGH)^T \right)^{-1}, \Phi = e^{A_c T} \quad (3.60)$$

and

$$f_{Nu} = \begin{bmatrix} A_N A_{N-1} \cdots A_2 B_1 & A_N A_{N-1} \cdots A_3 B_2 & \cdots & B_N \end{bmatrix} \begin{bmatrix} u_r(t_{r_0}) \\ u_r(t_{r_1}) \\ \vdots \\ u_r(t_{r_{N-1}}) \end{bmatrix} \quad (3.61)$$

do not depend on $x(t_0)$.

Therefore the constraint (3.49) becomes

$$HF\hat{\theta} \geq \delta t_{\min} - HM [x_r(t_{r_N}) - f_{Nu}] + HM\Phi x(t_0) \quad (3.62)$$

i.e.

$$[-HF \ HM\Phi] \begin{bmatrix} \hat{\theta} \\ x(t_0) \end{bmatrix} \leq HM[x_r(t_{r_N}) - f_{N_u}] - \delta t_{min} \quad (3.63)$$

The inequality (3.63) describes a polyhedron $\mathcal{P} \subset \mathbb{R}^{N-1}$. The feasibility region $\mathcal{X}_F$ corresponds to the set of points $x(t_0)$ for which there exist $\hat{\theta}$ such that

$$\begin{bmatrix} \hat{\theta} \\ x(t_0) \end{bmatrix} \in \mathcal{P} \quad (3.64)$$

Therefore, $\mathcal{X}_F$ can be obtained by projecting $\mathcal{P}$ onto the subspace corresponding to the $x(t_0)$ components. In what follows, the Multi-Parametric Toolbox developed by Herceg *et al.* (2013) was used for this purpose.

3.4 Application Examples

3.4.1 Double integrator

This example is concerned with the double integrator model (2.50), (2.51) presented in Section 2.3. The oscillation period is $T = 6$ s, and comprises $N = 4$ switching times, with $L_{on} = 2$ and $L_{off} = 1$.

Fig. 3.2 (a) and Fig. 3.2 (b) present the target trajectory $x_r(t)$ and the associated input signal $u_r(t)$. It should be noted that the target trajectory is open-loop unstable, in the sense that the state drifts away if no perturbations on the switching times are applied. This is the case of the trajectory depicted in Fig. 3.2 (a).

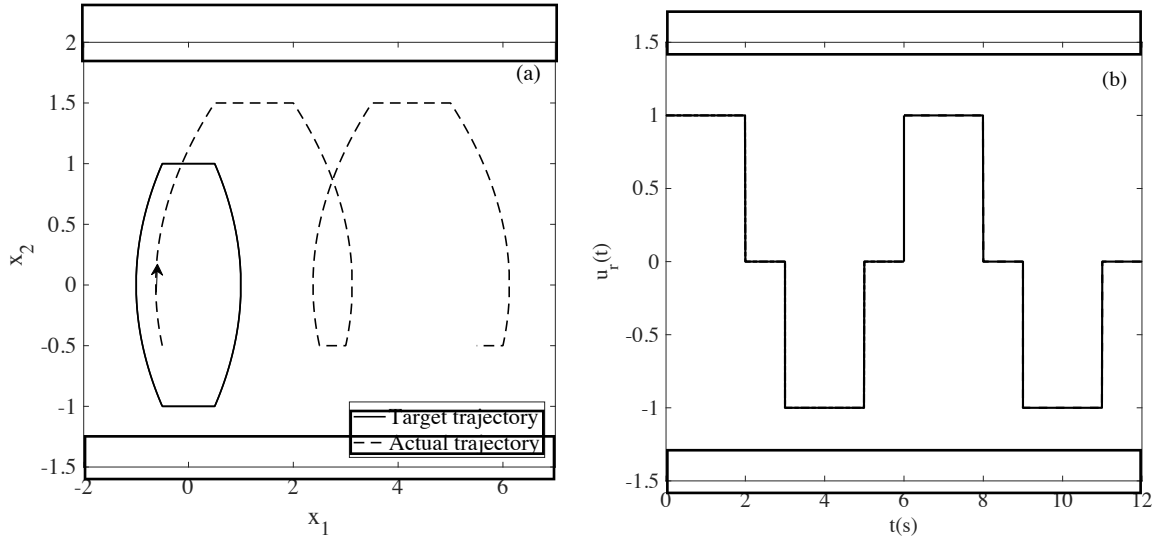


FIGURE 3.2 – Simulation results for $\delta t_k = 0$: (a) State trajectory and (b) input signal.

The closed loop control law (developed in Section 3.3) will be aimed at stabilizing the reference trajectory, so that it attracts trajectories initiated in its vicinity.

Initially, the switching-time constraints were not enforced, i.e. $\Delta t_{on} = 0$ and $\Delta t_{off} = 0$. Figure 3.3 presents the feasibility regions for the four possible target points in the target trajectory, as defined in Section 3.3.4.

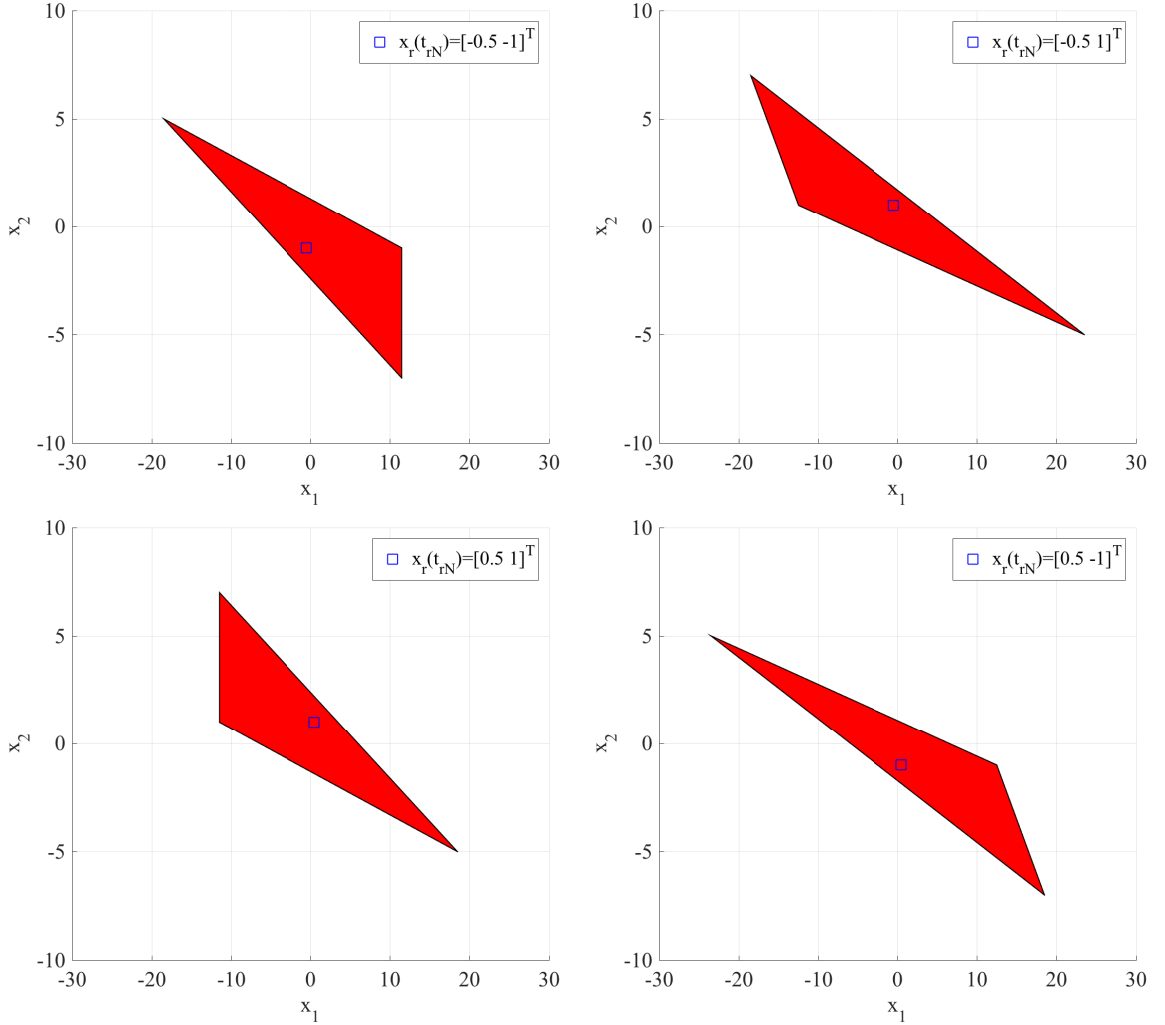


FIGURE 3.3 – Feasibility regions for different target points with $\Delta t_{on} = 0$ and $\Delta t_{off} = 0$.

If the initial condition is in one of these regions, then it always will be possible to choose a target point in order to obtain a feasible problem. Therefore, the union of these regions is the complete feasibility region, as shown in Fig. 3.4.

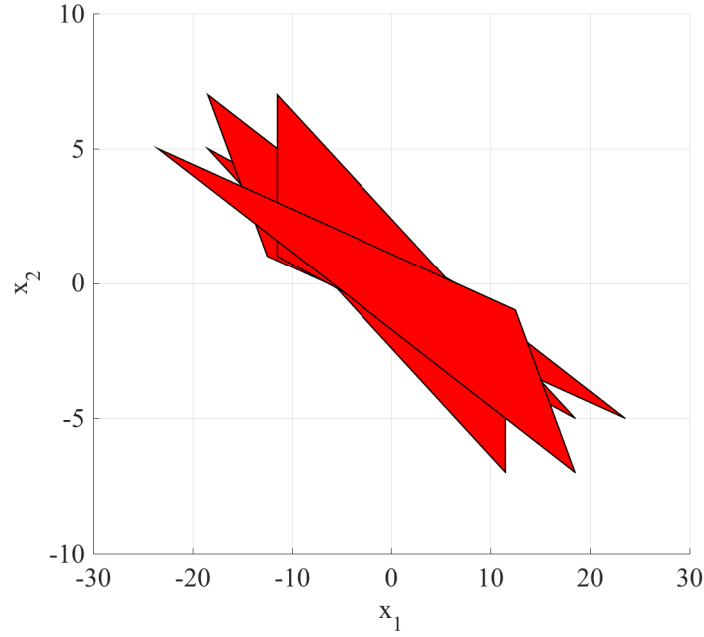


FIGURE 3.4 – Combined plot of the feasibility regions presented in Figure 3.3.

The weights of the cost function (3.24) were chosen as $\rho = 0.01$, $Q = I$, and the final target point was set to $x_r(t_{r_N}) = [-0.5 - 1]^T$. All results presented herein were obtained by using the quadratic programming solver of the Matlab Optimization ToolboxTM, with the interior-point-convex algorithm option.

Figure 3.5 presents the results obtained with an initial state $x(t_0) = [-1.5 - 1]^T$, which is located outside the target trajectory. The inequality constraints (3.34) were imposed, with $\Delta t_{on} = 0$ and $\Delta t_{off} = 0$. As can be seen in Figure 3.5 (b), the final target point was not reached at the end of the first horizon, even though the terminal constraint (3.33) was enforced in the optimization procedure. This outcome can be ascribed to a mismatch between the predicted trajectory and the actual trajectory, resulting from the approximations (3.9) and (3.11) employed in the linearization procedure described in Section 3.2. However, after the second horizon the difference with respect to the target became negligible.

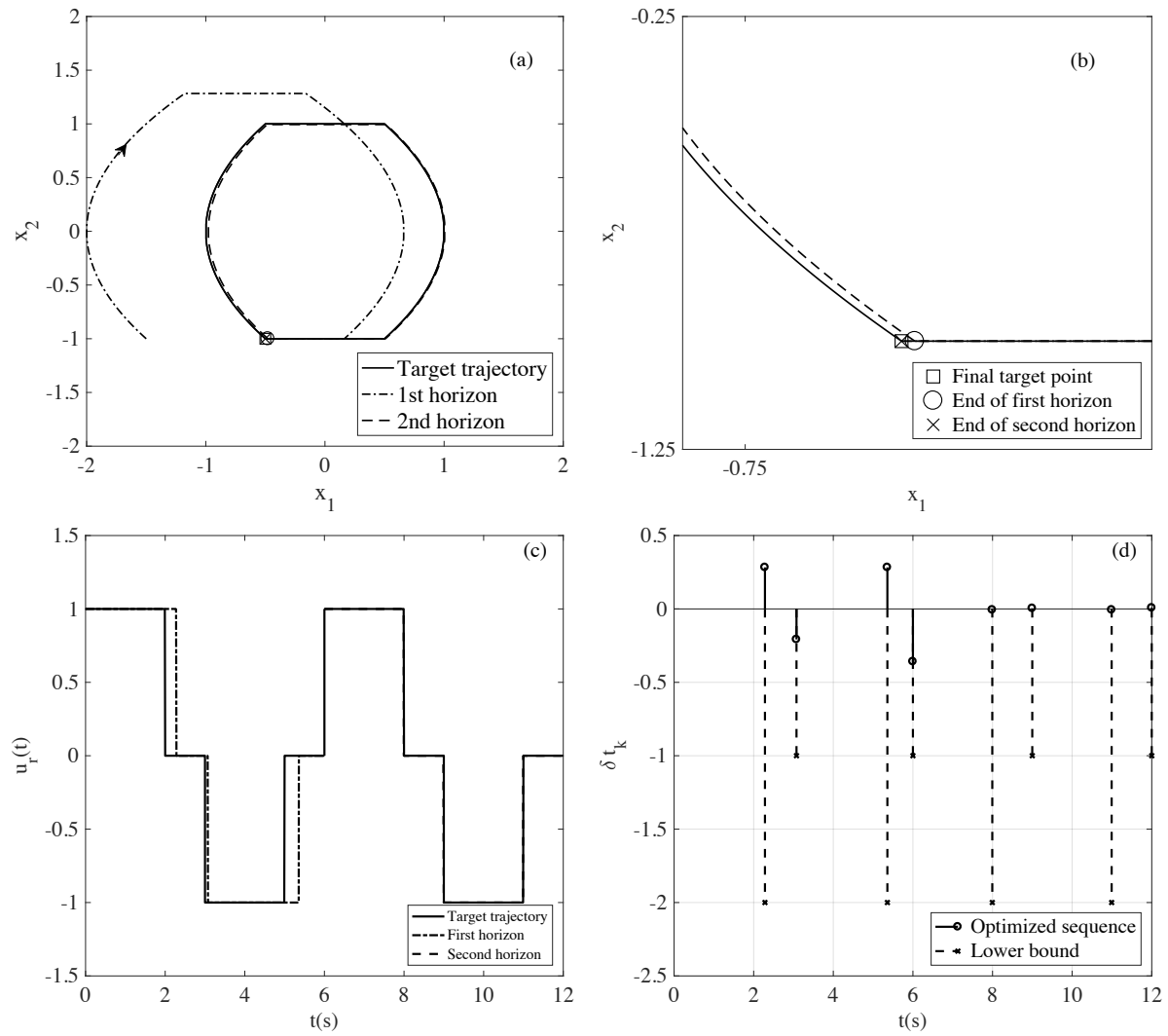


FIGURE 3.5 – Initial condition outside the target trajectory: (a) state trajectory, (b) enlarged view around the final target point, (c) control signal, (d) optimized sequence of perturbations δt_k .

Similar results were obtained with an initial condition $x(t_0) = [-0.5 \ -0.5]^T$, inside the target trajectory, as shown in Figure 3.6.

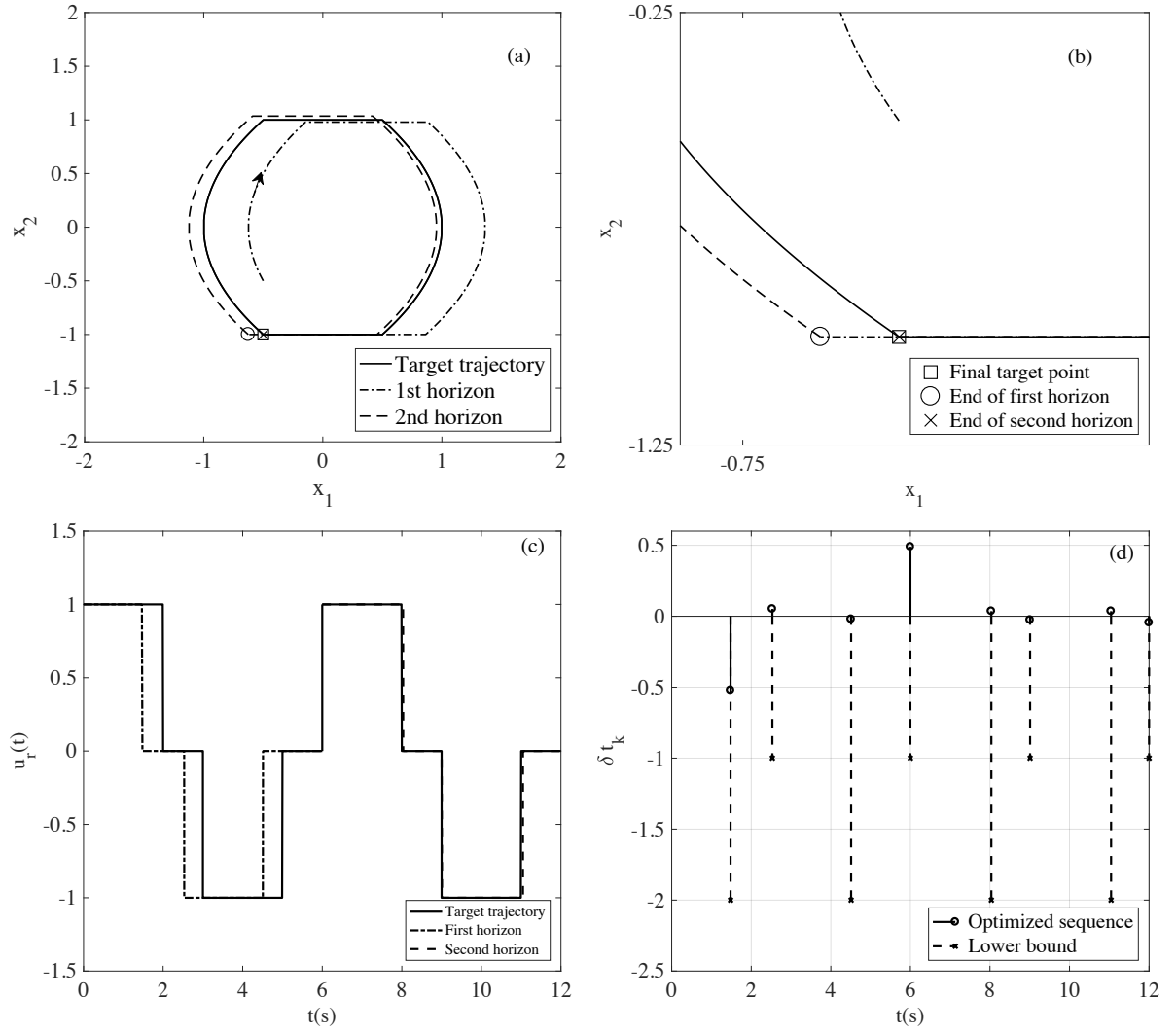


FIGURE 3.6 – Initial condition inside the target trajectory: (a) state trajectory, (b) enlarged view around the final target point, (c) control signal, (d) optimized sequence of perturbations δt_k .

To illustrate the possibility of enforcing switching-time constraints the inequality (3.34) was imposed with $\Delta t_{on} = 1.625$ and $\Delta t_{off} = 0.625$ in (3.35). The feasibility regions for different target points are presented in Figure 3.7. A combined plot of the four regions is shown in Fig. 3.8. As can be seen, they are considerably smaller compared to the region in Fig. 3.4, if the axis scales are taken into account.

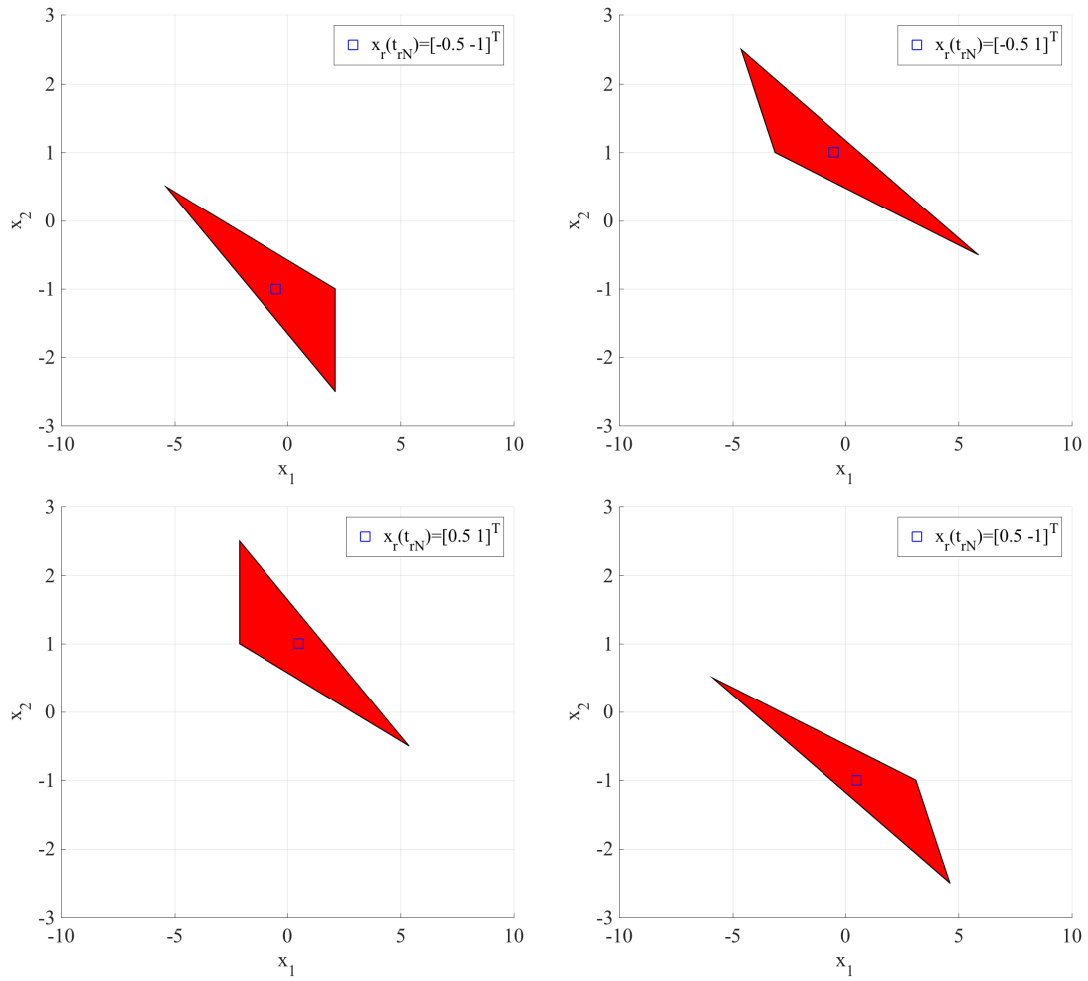


FIGURE 3.7 – Feasibility regions for different target points, with $\Delta t_{on} = 1.625$ and $\Delta t_{off} = 0.625$.

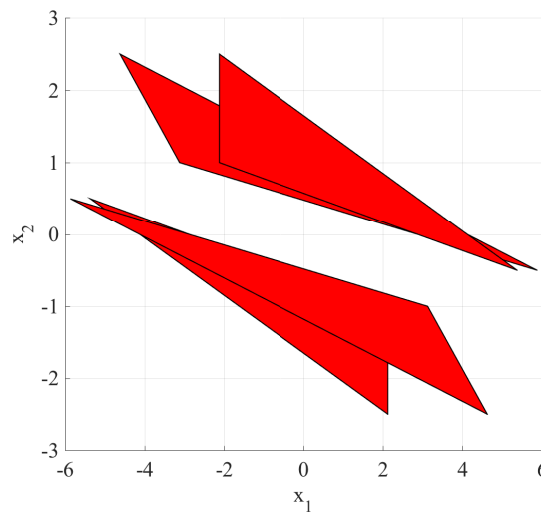


FIGURE 3.8 – Combined plot of the feasibility regions presented in Figure 3.7.

The closed-loop simulation starting from $x(t_0) = [-0.5 \ -0.5]^T$ was repeated with the inclusion of the switching-time constraints. More specifically, the lower bounds were calculated as in (3.35) with $\Delta t_{on} = 1.625$ and $\Delta t_{off} = 0.625$, resulting in $\delta t_{min} = [-0.375 \ -0.375 \ -0.375]^T$. The results are shown in Figure 3.9. As can be seen in Figure 3.9 (d), the constraints were properly enforced. Again, the state trajectory was driven to the target, with negligible error after the second optimization horizon.

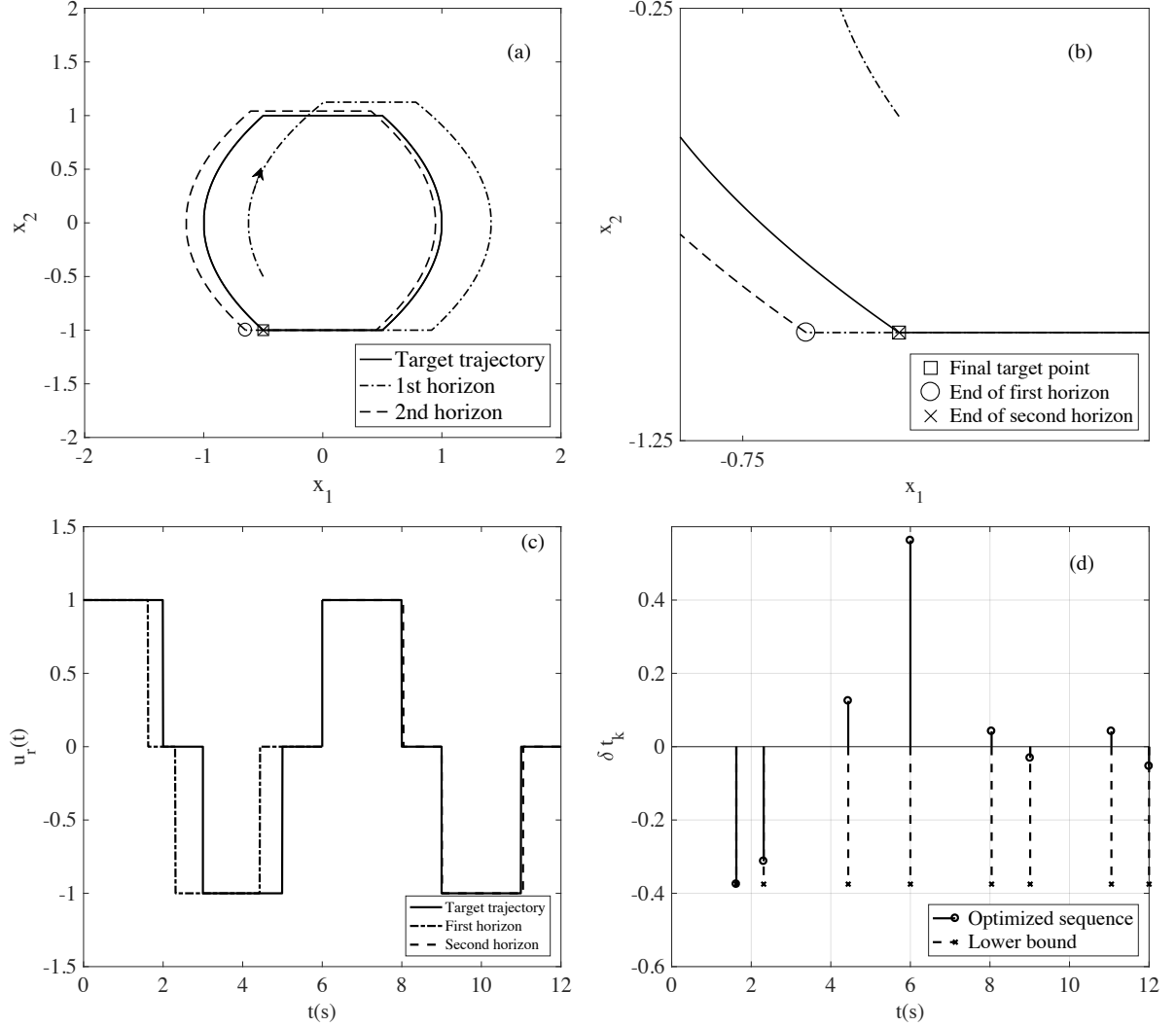


FIGURE 3.9 – Simulation results in the presence of switching-time constraints: (a) state trajectory, (b) enlarged view around the final target point, (c) control signal, (d) optimized sequence of perturbations δt_k .

As an additional example, consider the target trajectory (h) in Figure 2.6, which has an oscillation period $T = 10$ s and comprises $N = 4$ switching times. This is one of the two trajectories with smallest amplitude along x_1 axis, as shown in Figure 2.7.

Figure 3.10 presents the feasibility regions obtained for the four possible target points in the target trajectory, in the case where the switching-time constraints are not imposed

(i.e. with $\Delta t_{on} = 0$ and $\Delta t_{off} = 0$). A combined plot of the four regions is shown in Figure 3.11.

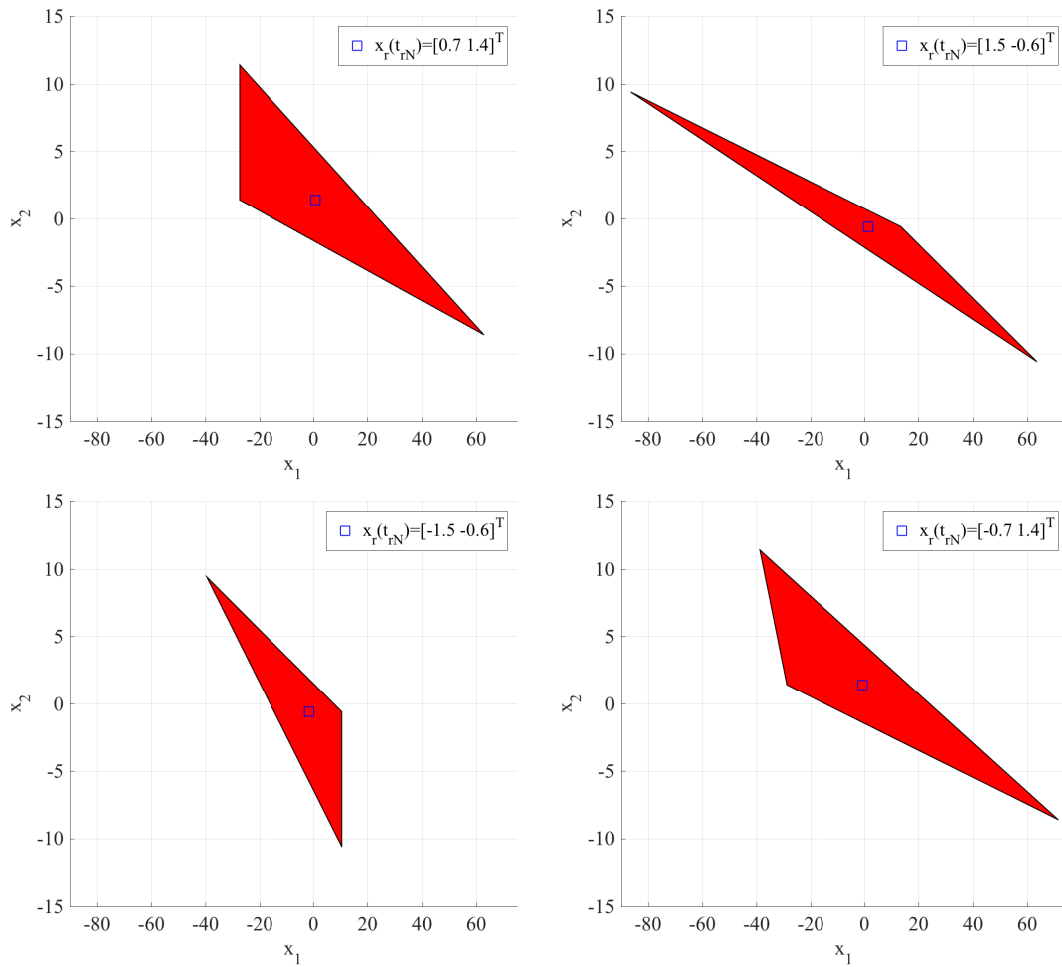


FIGURE 3.10 – Feasibility regions for different target points with $\Delta t_{on} = 0$ and $\Delta t_{off} = 0$.

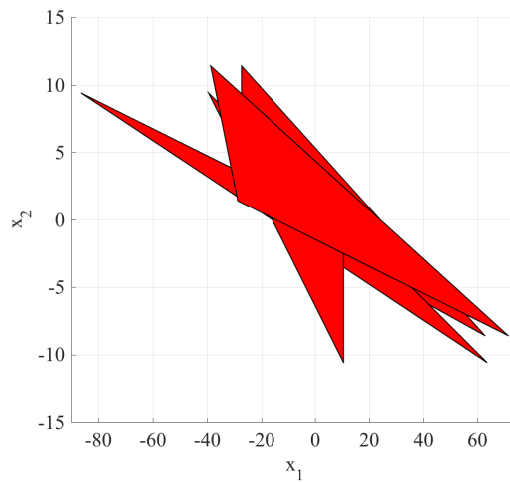


FIGURE 3.11 – Combined plot of the feasibility regions presented in Figure 3.10.

The weights of the cost function (3.24) were chosen as $\rho = 0.01$, $Q = I$, as in the preceding example, and the final target point was set to $x_r(t_{r_N}) = [0.7 \ 1.4]^T$. Figure 3.12 presents the results obtained with an initial state $x(t_0) = [-10 \ 2]^T$, located outside the target trajectory. As can be seen in Figure 3.12 (b), the final target point was not reached at the end of the first horizon, even though the terminal constraint (3.33) was enforced in the optimization procedure. As previously mentioned, this outcome can be ascribed to a mismatch between the predicted trajectory and the actual trajectory. However, after the second horizon the difference with respect to the target became negligible.

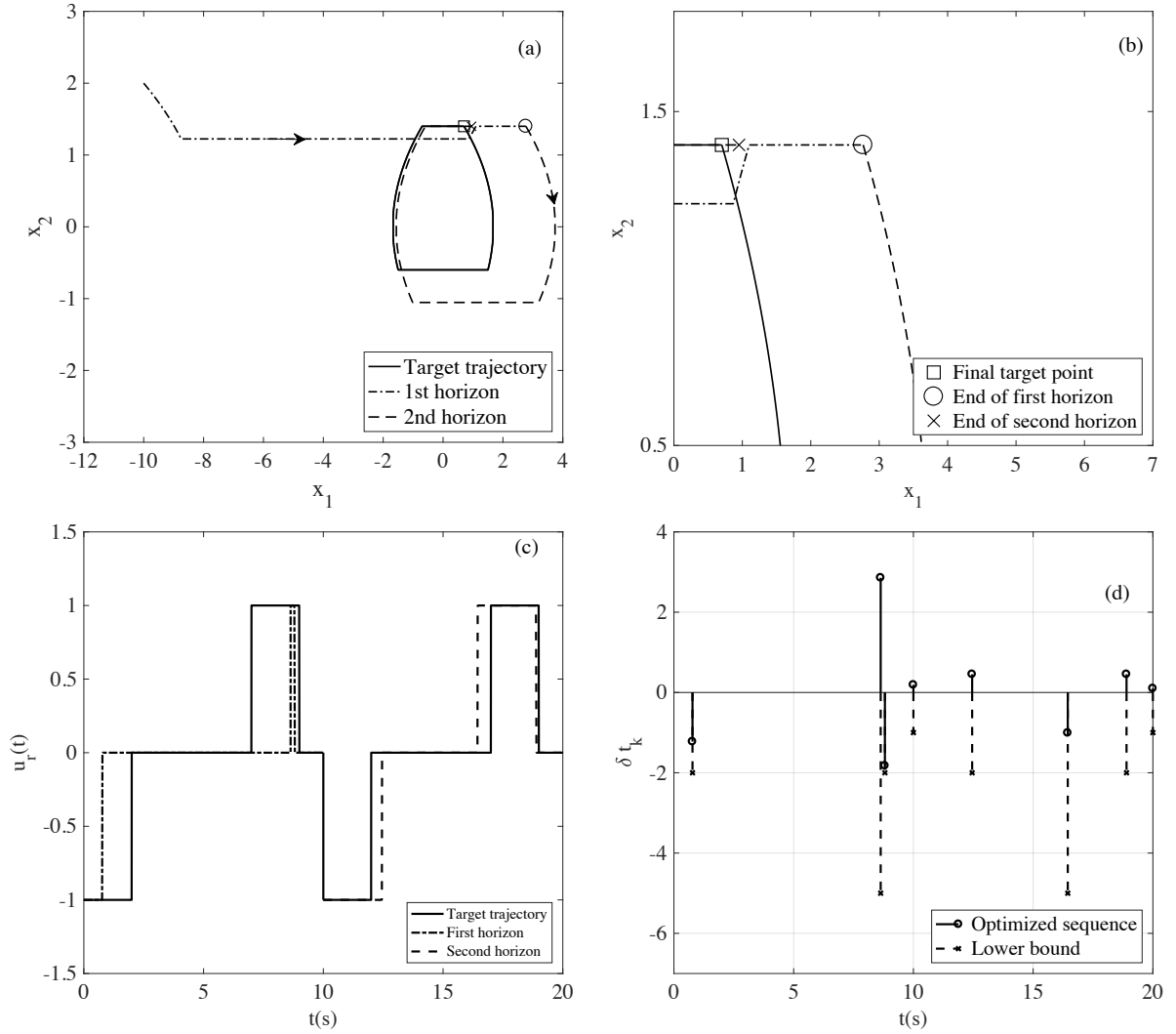


FIGURE 3.12 – Simulation results: (a) state trajectory, (b) enlarged view around the final target point, (c) control signal, (d) optimized sequence of perturbations δt_k .

For the next case the switching-time constraints were imposed by taking into account the constraint (3.34), with $\Delta t_{on} = 1.625$ and $\Delta t_{off} = 0.625$. The feasibility regions associated to the four possible target points are presented in Figures 3.13 and 3.14.

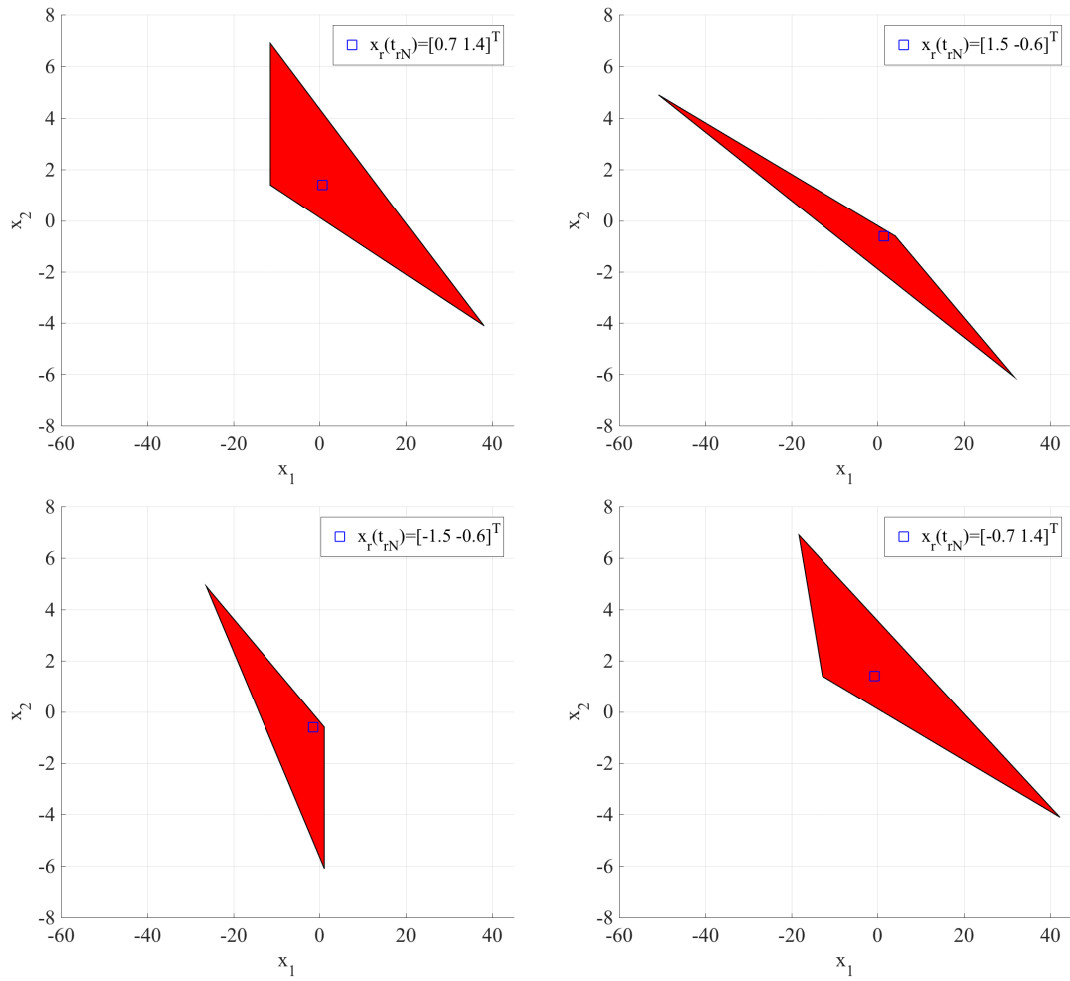


FIGURE 3.13 – Feasibility regions for different target points, with $\Delta t_{on} = 1.625$ and $\Delta t_{off} = 0.625$.

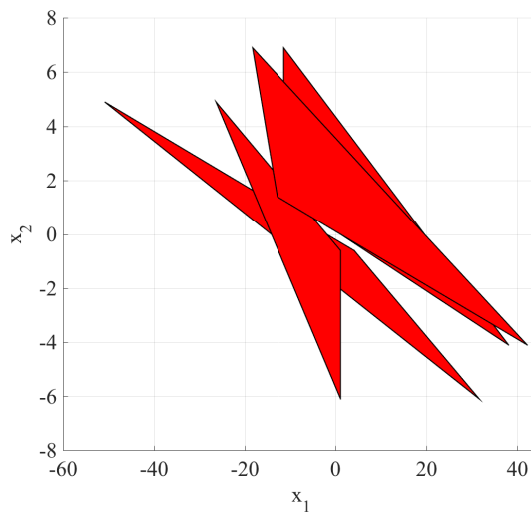


FIGURE 3.14 – Combined plot of the feasibility regions presented in Figure 3.13.

The closed-loop simulation starting from $x(t_0) = [-10 \ 2]^T$ was repeated with the inclusion of the switching-time constraints. More specifically, the lower bounds were calculated as in (3.35) with $\Delta t_{on} = 1.625$ and $\Delta t_{off} = 0.625$, resulting in $\delta t_{min} = [-0.375 - 4.375 - 0.375 - 0.375]^T$. The results are shown in Figure 3.15. As can be seen in Figure 3.15 (d), the constraints were properly enforced. Again, the state trajectory was driven to the target, with negligible error after the second optimization horizon.

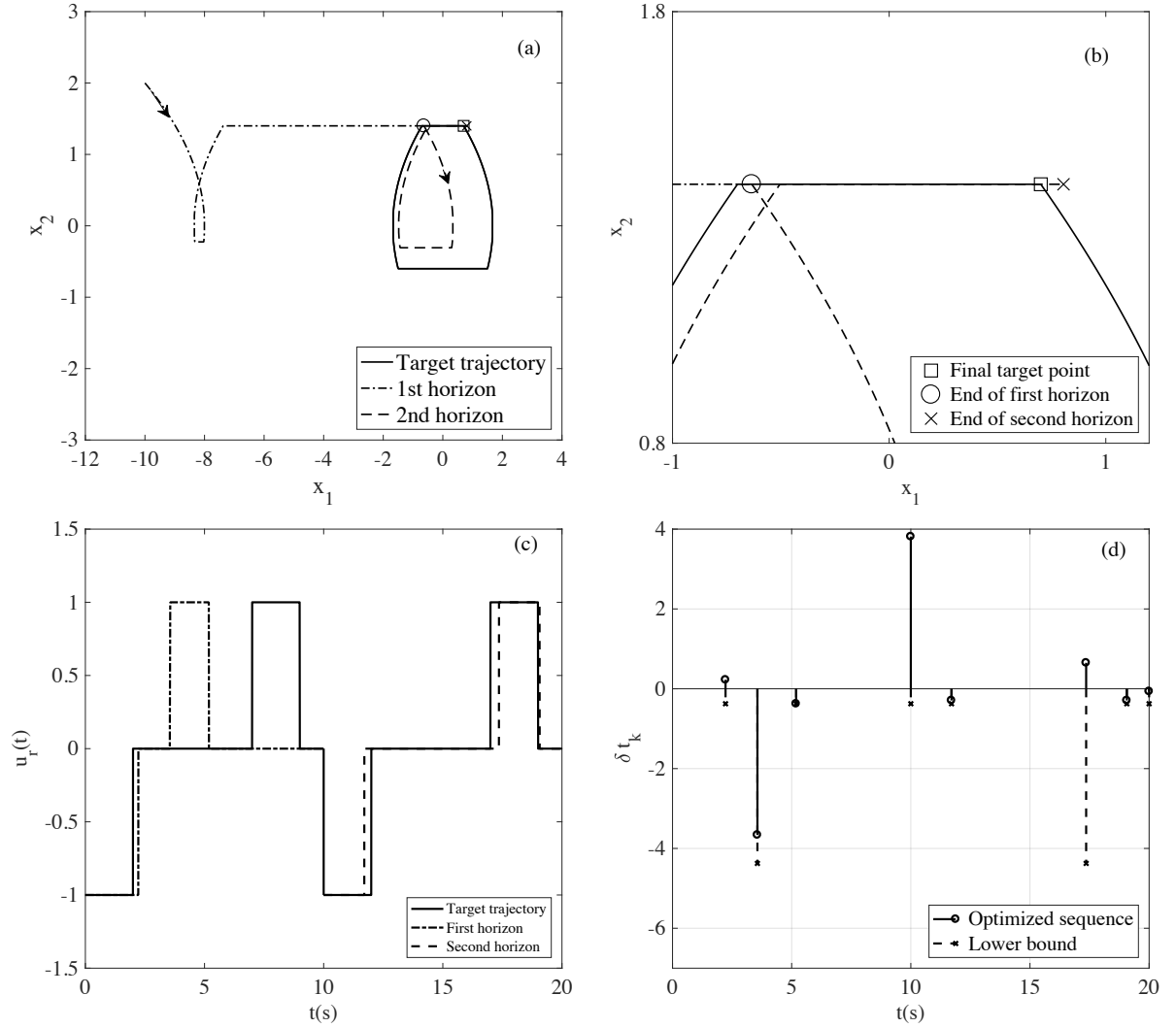


FIGURE 3.15 – Simulation results in the presence of switching-time constraints: (a) state trajectory, (b) enlarged view around the final target point, (c) control signal, (d) optimized sequence of perturbations δt_k .

3.4.2 Two-disk-spring system

This example is concerned with the two-disk-spring system described in Section 2.3.2. It is assumed that the state should be driven to the target trajectory (h) in Figure 2.9, which has an oscillation period $T = 10s$, and comprises $N = 4$ switching times. Herein

the optimization horizon will be extended to two oscillation periods due to the increased number of state variables (four) involved in the terminal constraint (3.33). The weights of the cost function (3.24) were chosen as $\rho = 0.01$, $Q = I$, and the final target point was set to $x_r(t_{r_N}) = [0.1536 \ 0.017 \ 0.3061 \ -0.0327]^T$.

Figures 3.16 and 3.17 presents the results obtained with an initial state $x(t_0) = [0.5 \ 0.1 \ -0.001 \ 0]^T$, the switching-time constraints are not imposed (i.e. with $\Delta t_{on} = 0$ and $\Delta t_{off} = 0$). The resulting closed-loop trajectories are presented in Figure 3.16. The input control signal, and the optimized sequence of perturbations are illustrated in Figure 3.17 (a) and (b), respectively. As can be seen, the state and control trajectories converge to the target profiles, with small error at the end of the third horizon.

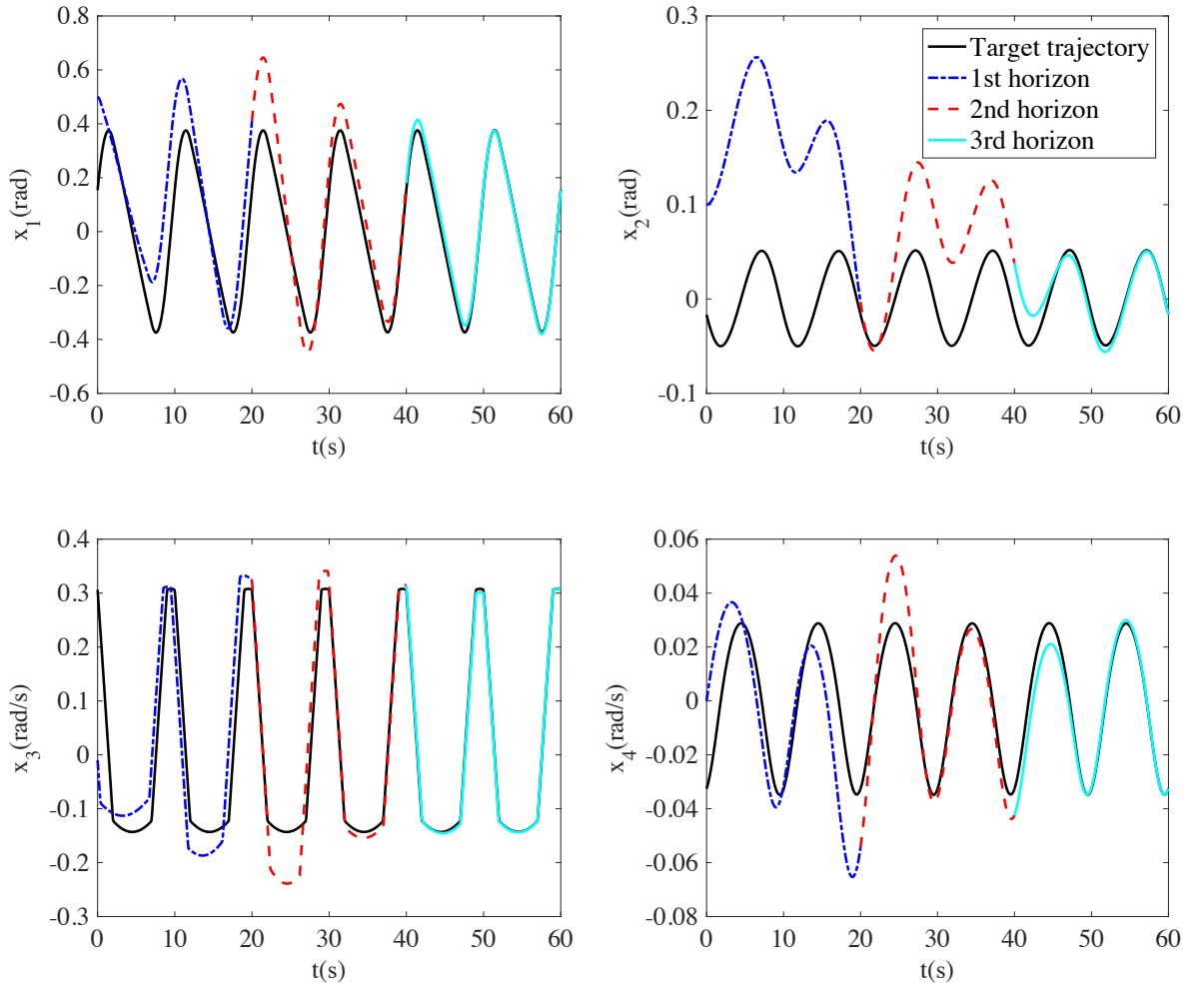
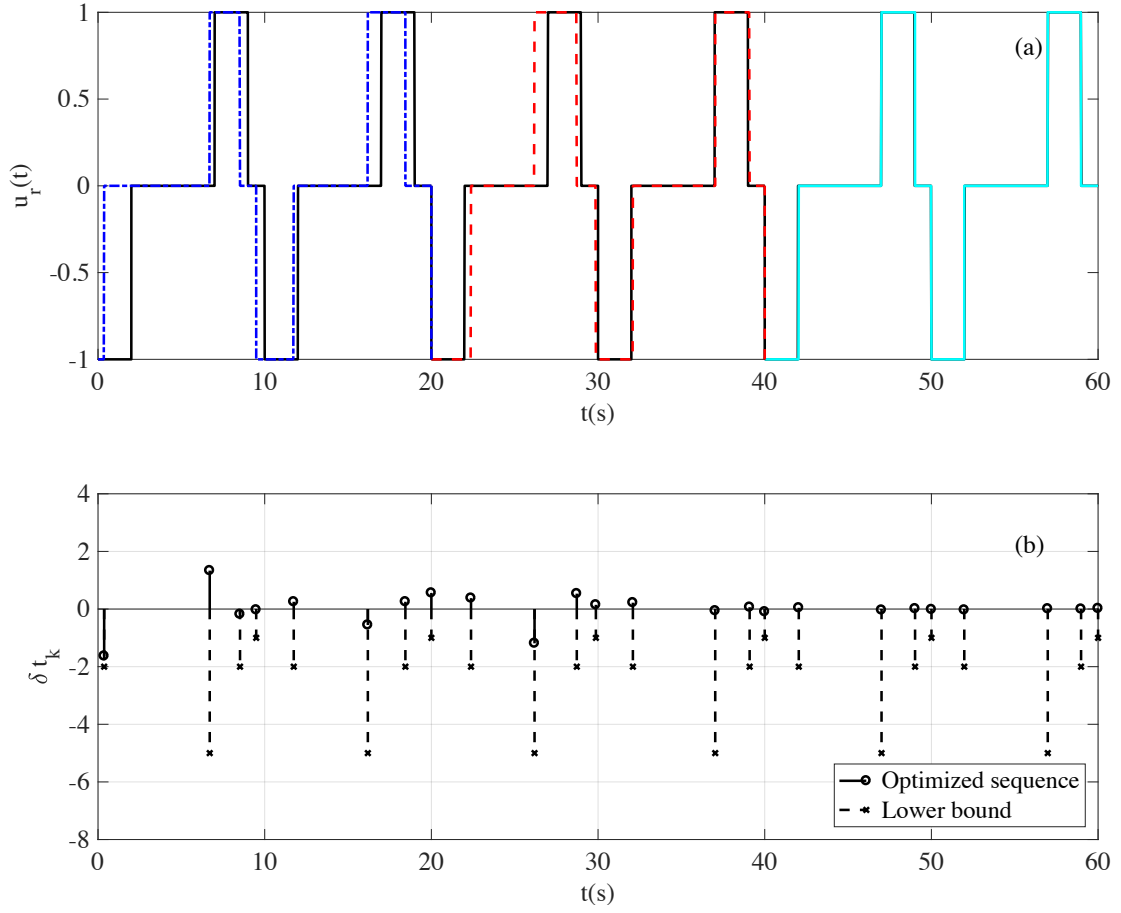


FIGURE 3.16 – State trajectories.


 FIGURE 3.17 – (a) Control signal, (b) optimized sequence of perturbations δt_k .

The next simulation illustrates the imposition of switching-time constraints. In this case, the lower bounds were set to $\Delta t_{on} = 1.625$ and $\Delta t_{off} = 0.625$ in (3.35), which resulted in $\delta \mathbf{t}_{min} = [-0.375 \ -4.375 \ -0.375 \ -0.375 \ -0.375 \ -4.375 \ -0.375 \ -0.375]^T$, as defined in (3.50). The results are shown in Figures 3.18 and 3.19. The constraint was properly enforced in the optimization procedure as illustrated in Figure 3.19 (b). Again, the results indicates convergence to the reference trajectory, although with larger error during the first two horizons compared to Figure 3.16.

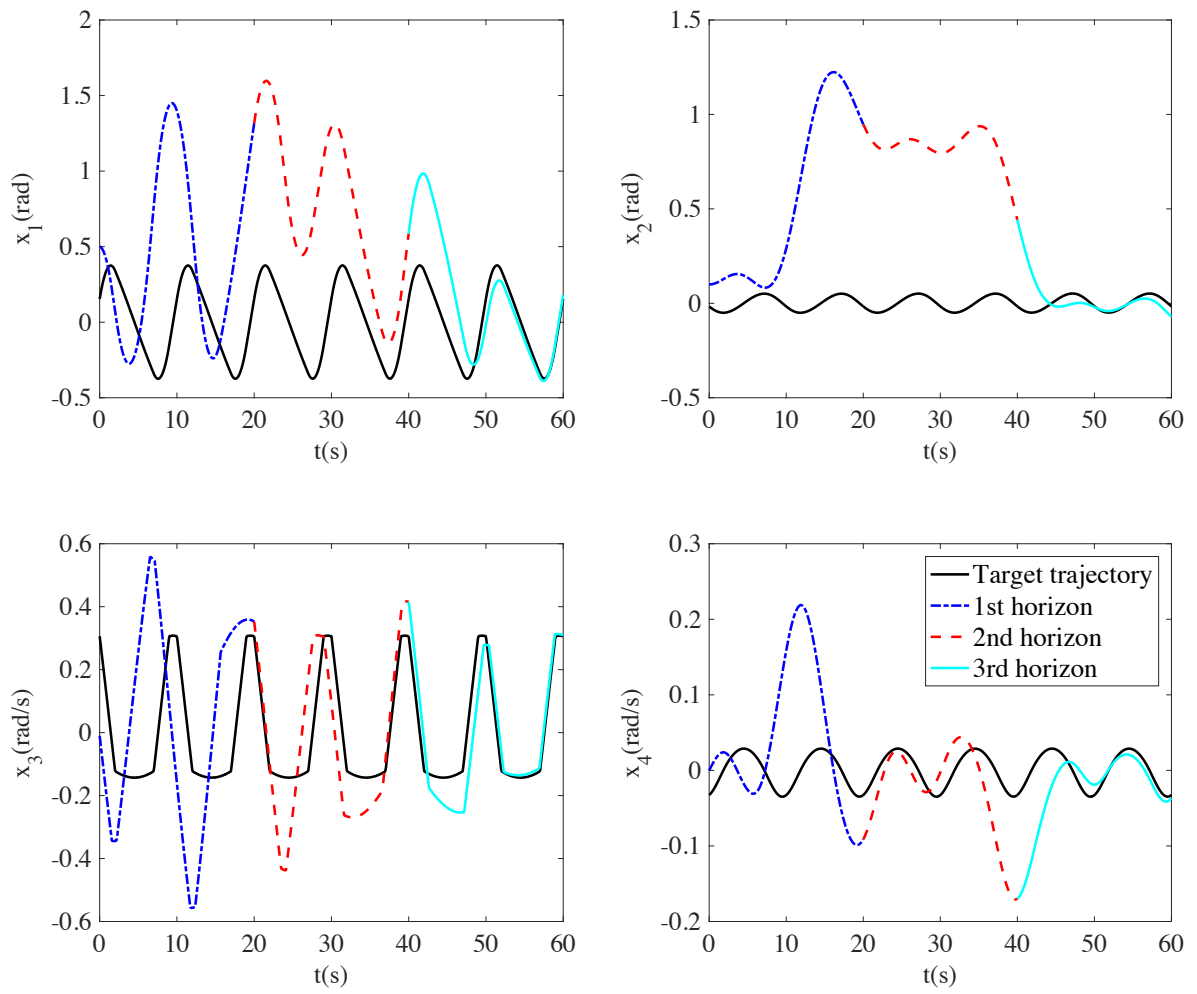
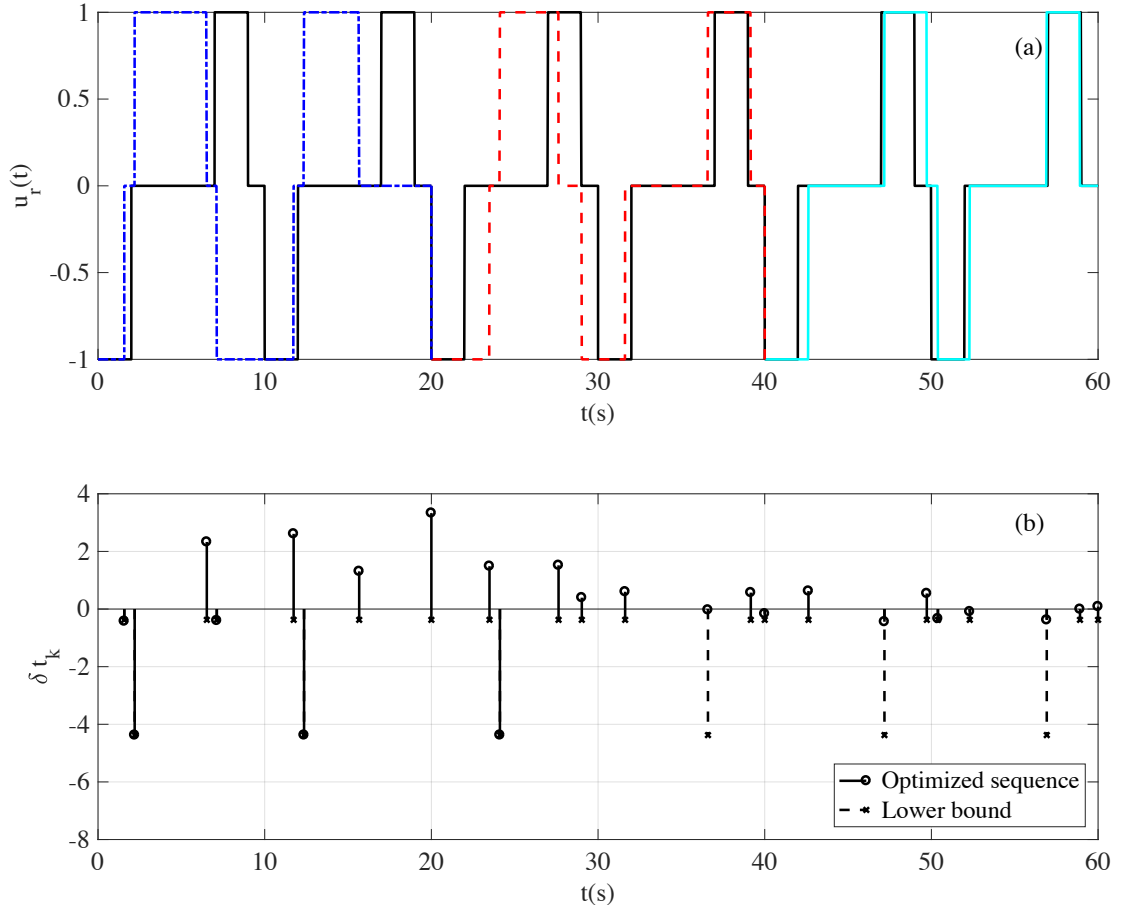


FIGURE 3.18 – State trajectories.

FIGURE 3.19 – (a) Control signal, (b) optimized sequence of perturbations δt_k .

As discussed in the previous example, the deviation from the target point at the end of each optimization horizon can be ascribed to the mismatch between the predicted and actual state trajectories, stemming from the approximations (3.9) and (3.11). Arguably, the approximation errors would become less pronounced if the magnitude of the perturbations δt_k were reduced. Such a reduction can be achieved by increasing the design weight ρ . To investigate this possibility, the simulation was repeated after increasing this weight from $\rho = 0.01$ to $\rho = 1$. The results are presented in Figures 3.20 and 3.21. As can be seen by comparing Figures 3.18 and 3.20, the state deviation from the target became much smaller at the end of the second horizon. This improvement can be associated to the reduction in the magnitude of the perturbations δt_k in the beginning of the second optimization horizon, as can be noted by comparing Figures 3.19 and 3.21.

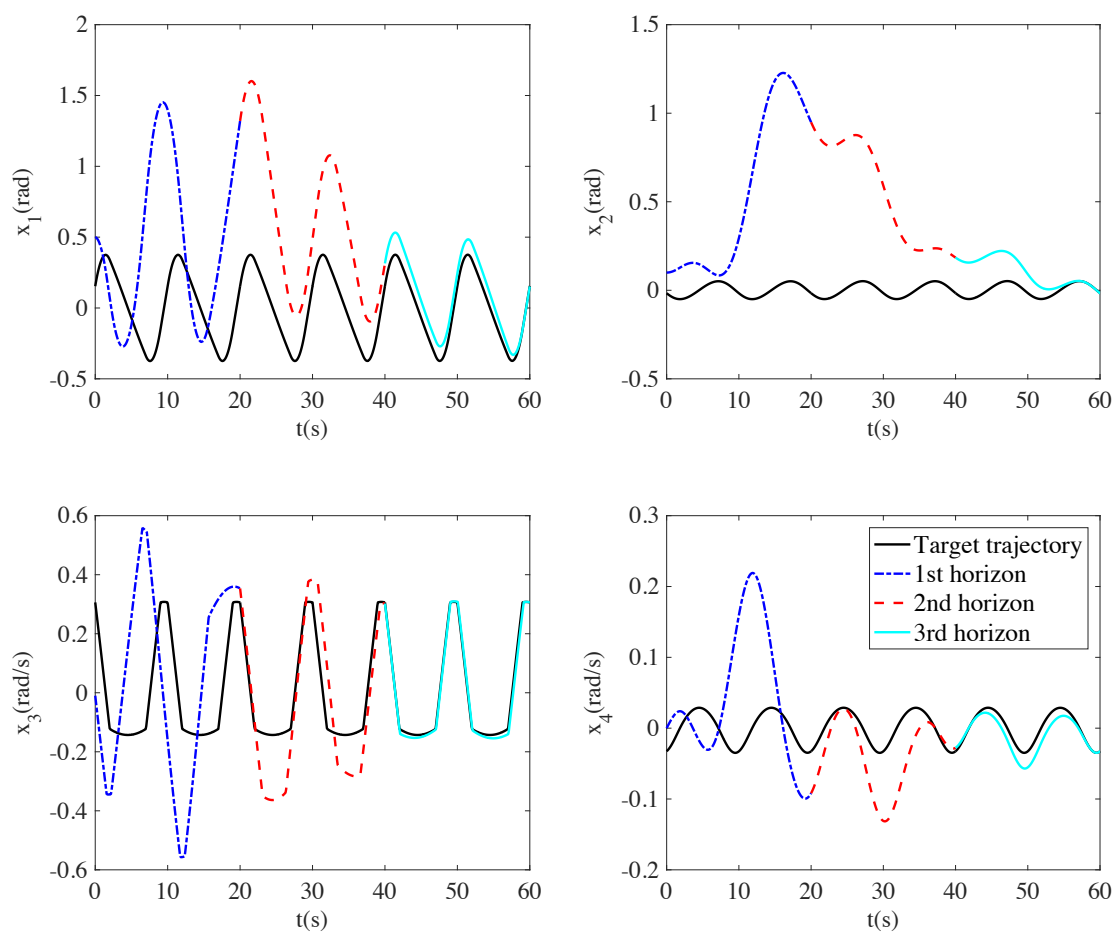
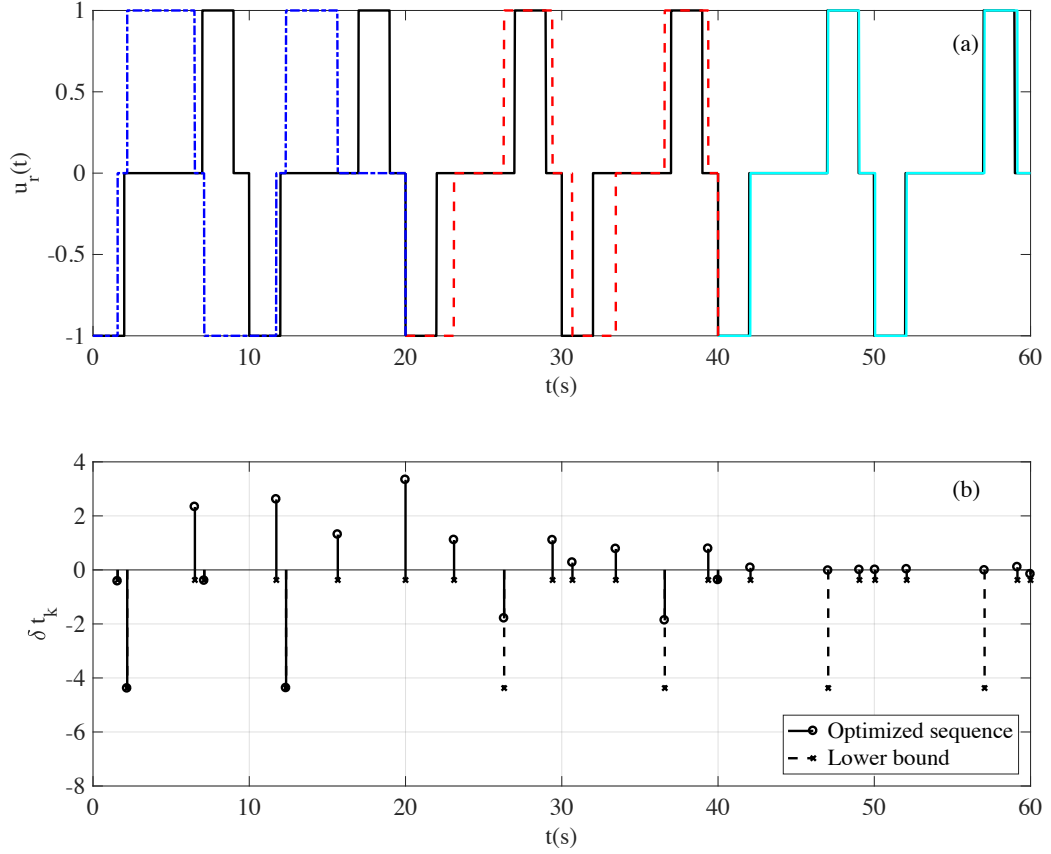


FIGURE 3.20 – State trajectories.

FIGURE 3.21 – (a) Control signal, (b) optimized sequence of perturbations δt_k .

3.5 Chapter conclusions

This chapter was concerned with a predictive control formulation developed to drive the state from an initial point to the target trajectory through perturbations in the switching times. An approximate model which describes the relationship between these perturbations and the resulting state trajectory was proposed. Appropriate perturbations were then determined through the solution of an optimal control problem. The proposed control law can only be used if the initial state point is feasible. Therefore, the characterization of the feasibility region for the initial state was also discussed.

Numerical examples using the double integrator and the two-disk-spring system were presented to illustrate the use of the proposed controller. It was noticed that, despite the enforcement of a terminal constraint in the optimization procedure, the actual state trajectory did not reach the target after the first horizon. This mismatch comes from the approximation employed in the linearization procedure. On the other hand, the difference between trajectories decreased significantly at the end of the second horizon for the double integrator, and at the end of the third horizon for the two-disk-spring system.

4 Conclusion

Summary of contributions

The first contribution of this work consisted in determining periodic state trajectories (addressed in Chapter 2) arising from the application of bidirectional on/off inputs in the form $\{-1, 0, +1\}$ subject to switching-time constraints. A suitable trajectory can then be selected by using engineering criteria.

In the second contribution, this trajectory was used as a target in a predictive control formulation. For this purpose, a linearization procedure was derived to describe the relation between perturbations in the switching times and the resulting state trajectory. These perturbations were used as decision variables in an optimal control problem. The characterization of the feasibility region for the initial state through computational geometry operations was also discussed.

Examples

The proposed method for determination of feasible periodic trajectories was illustrated with the use of a double integrator and a two-disk-spring model. The effect of the switching-time constraints was mainly related to a reduction in the number of feasible trajectories, as the constraints became more stringent. The two examples also illustrated the possibility of choosing a convenient trajectory in terms of the amplitude of the oscillations.

These examples were also employed in the second part of the thesis, which was concerned with the closed-loop control by means of optimized perturbations in the switching times. Even though a terminal constraint was enforced in the optimization procedure, the actual state trajectory displayed a deviation from the target at the end of the first horizon. This outcome can be ascribed to a mismatch between the predicted and actual trajectories, stemming from the approximations employed in the linearization procedure. However, the deviation became progressively smaller at the end of subsequence horizons, which resulted in a convergence of the state to the target trajectory.

Suggestions for future work

Future research could investigate the possibility of introducing an optimality criterion in the search for feasible trajectories, thus avoiding the need for the post-processing pro-

cedure employed in Chapter 2. Additionally, a robust controller could be designed to take into account the approximation errors in the linearization procedure, in order to obtain formal guarantees of convergence. Moreover, it could be of value to extend the contributions of the present work to systems with multiple input channels, as well as systems with more than three control levels (e.g. quantization levels in the output of a digital-to-analog converter). Control tasks involving switches between two or more subsystems could also be considered. Finally, validation studies in an experimental setting could be carried out.

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11. RESUMO: The present work conveys two contributions. Initially, from the application of bidirectional on/off input sequences subject to switching-time constraints, the periodic state trajectories of a discrete-time dynamic systems were determined. In this sense, the contribution of this work consists in characterizing feasible input sequences and associated state trajectories in terms of linear constraints involving integer variables. The second part of this work consists in the formulation of a predictive control law, making use of perturbations in the switching times, to drive the state from a given initial point to the target trajectory, properly selected by using some engineering criterion in the previous stage. For this reason a linearization procedure is derived to describe the relationship between the perturbations in the switching times and the state trajectory. To evaluate the proposed methods two numerical examples are presented. The first example concerns to a double integrator model. The second example brings a model of the double integrator with flexible mode. For each example a periodic trajectory used as a target is determined, and thus the control law is applied to drive the state to the desired target.			
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